Technical Training

Cooling System

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Training objectives

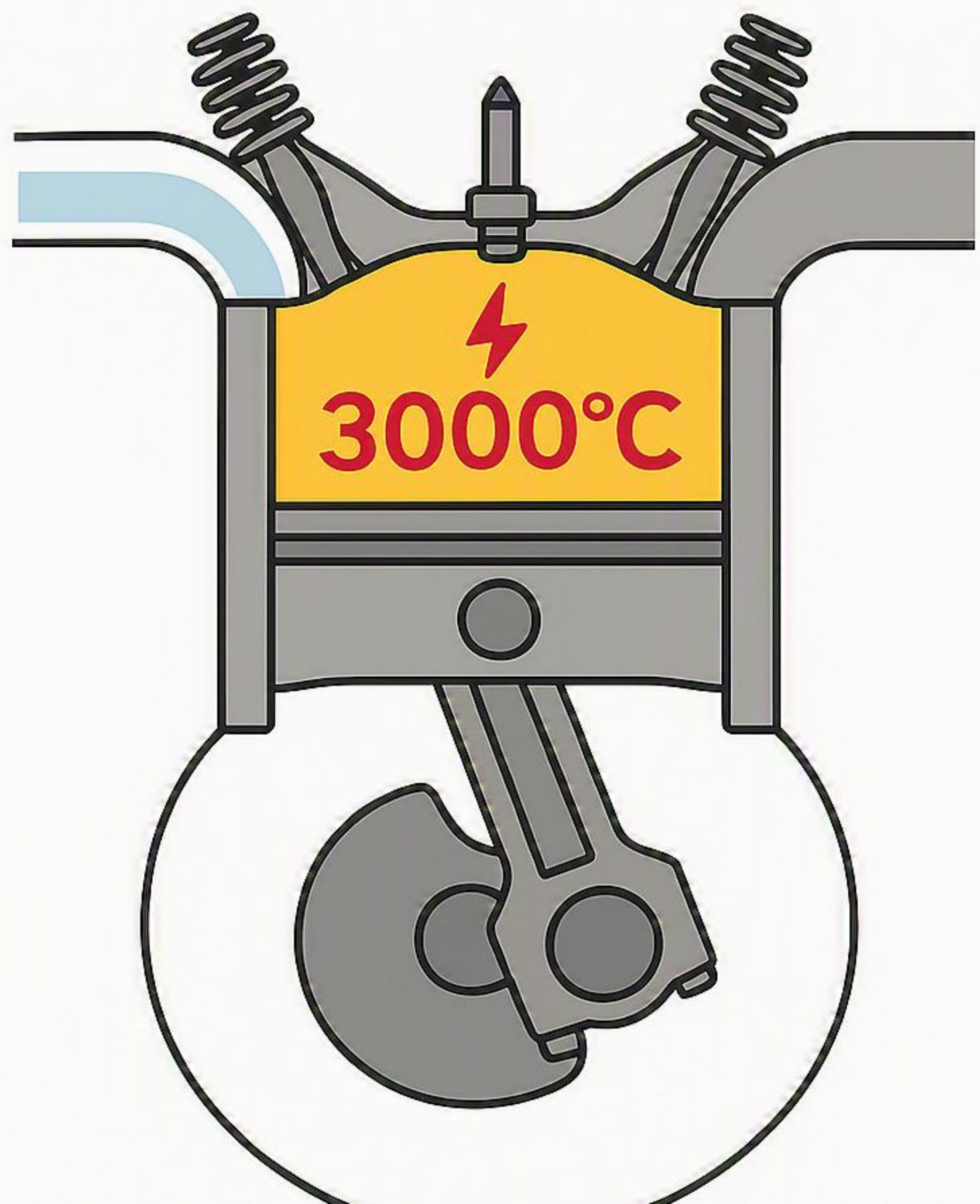
By the end of the day, you will be able to:

– Explain the different types of antifreeze and their properties
– Explain the components of a system fitted with an auxiliary pump
– Explain the cooling circuits of electric vehicles (EV)
– Explain the service and maintenance procedures for these systems

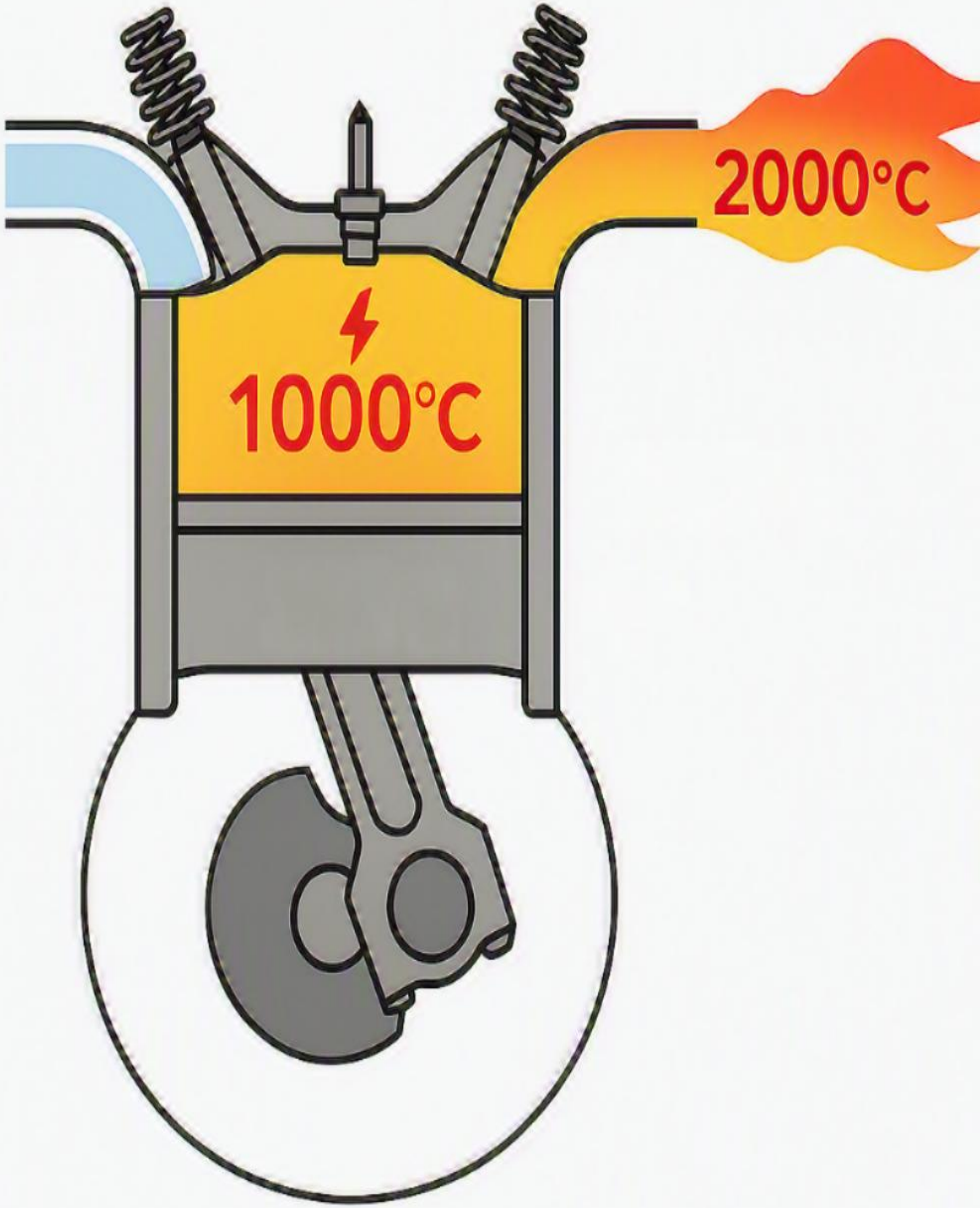
How do we handle this 3000 °C?

Where the heat goes

– 2/3 of the heat escapes through the exhaust
– The remaining 1/3 splits between the piston and the cylinder walls



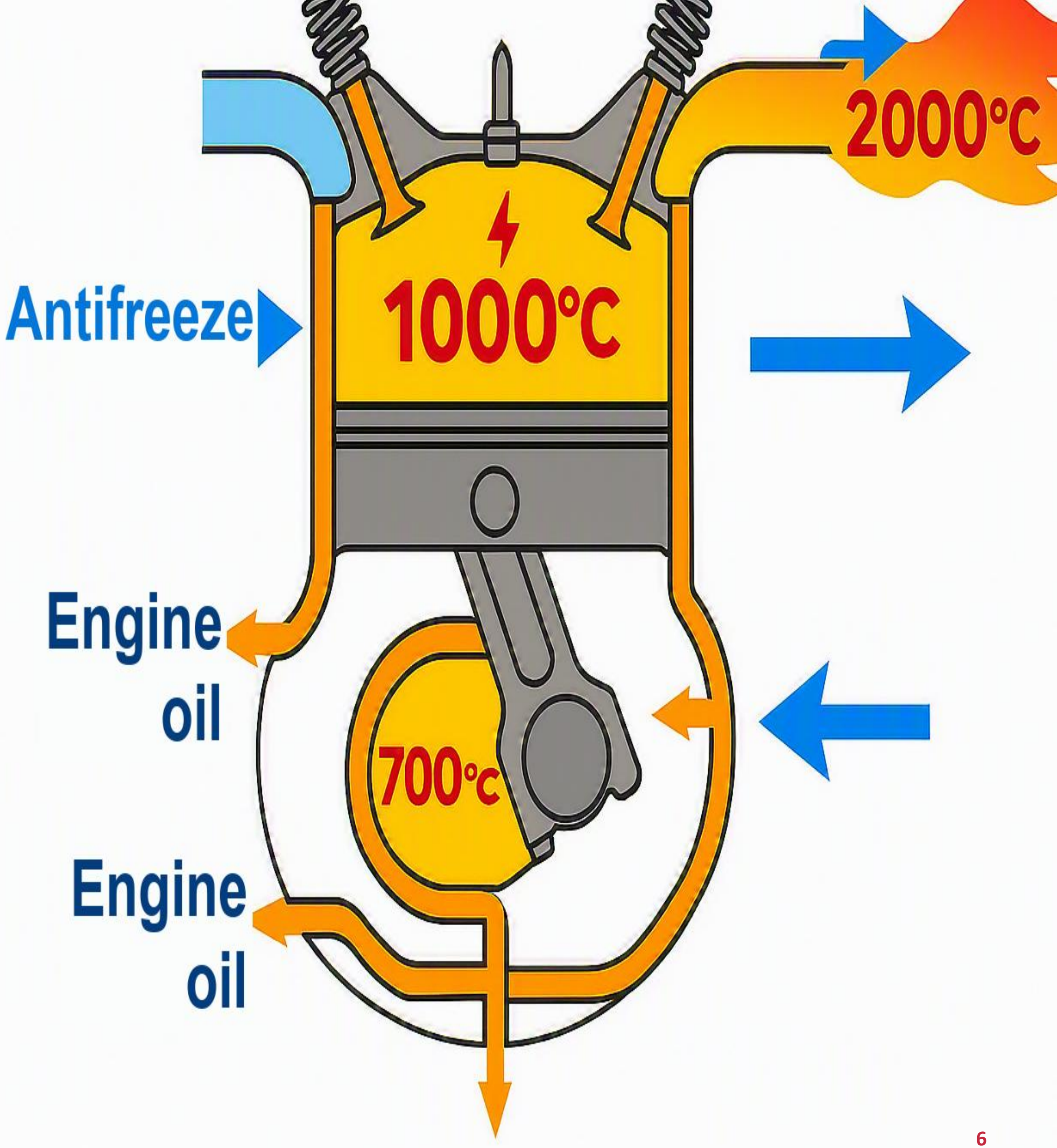
How do we handle the remaining 1000 °C?



Cooling system

Two fluids carry the heat away

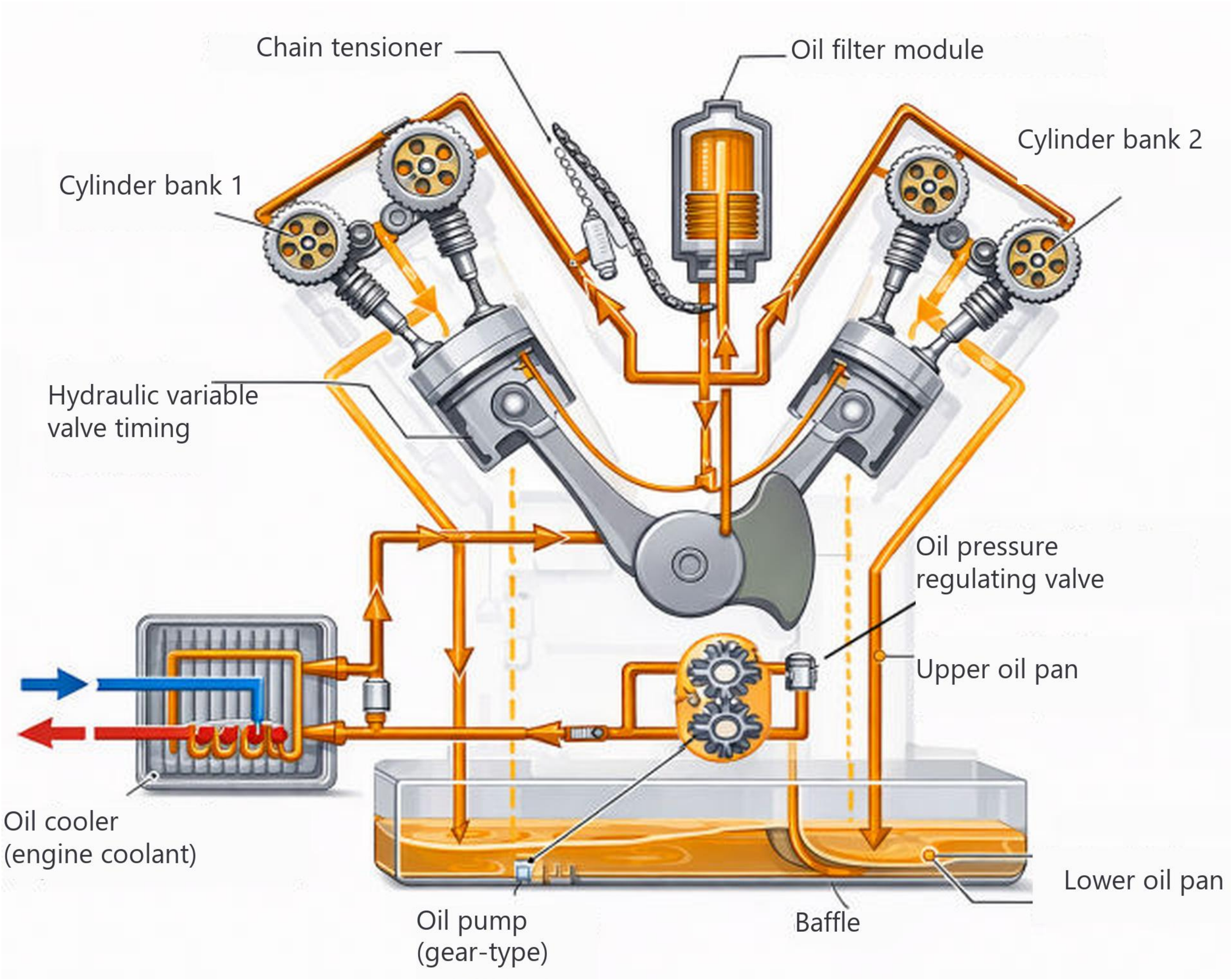
– Antifreeze
– Engine oil



The role of engine oil

Three roles

– Lubricate, to reduce wear
– Cool, mainly the piston
– Clean



Engine oil

The circuit and its six components

1. Oil pan
– Reservoir at the bottom of the engine, it holds the engine oil

2. Oil pump
– Draws oil from the pan and sends it under pressure throughout the engine

3. Oil filter
– Cleans the oil: particles, metal shavings, contaminants

4. Oil galleries
– Small internal passages in the engine block
– Distribute oil to the crankshaft, the camshaft and the pistons

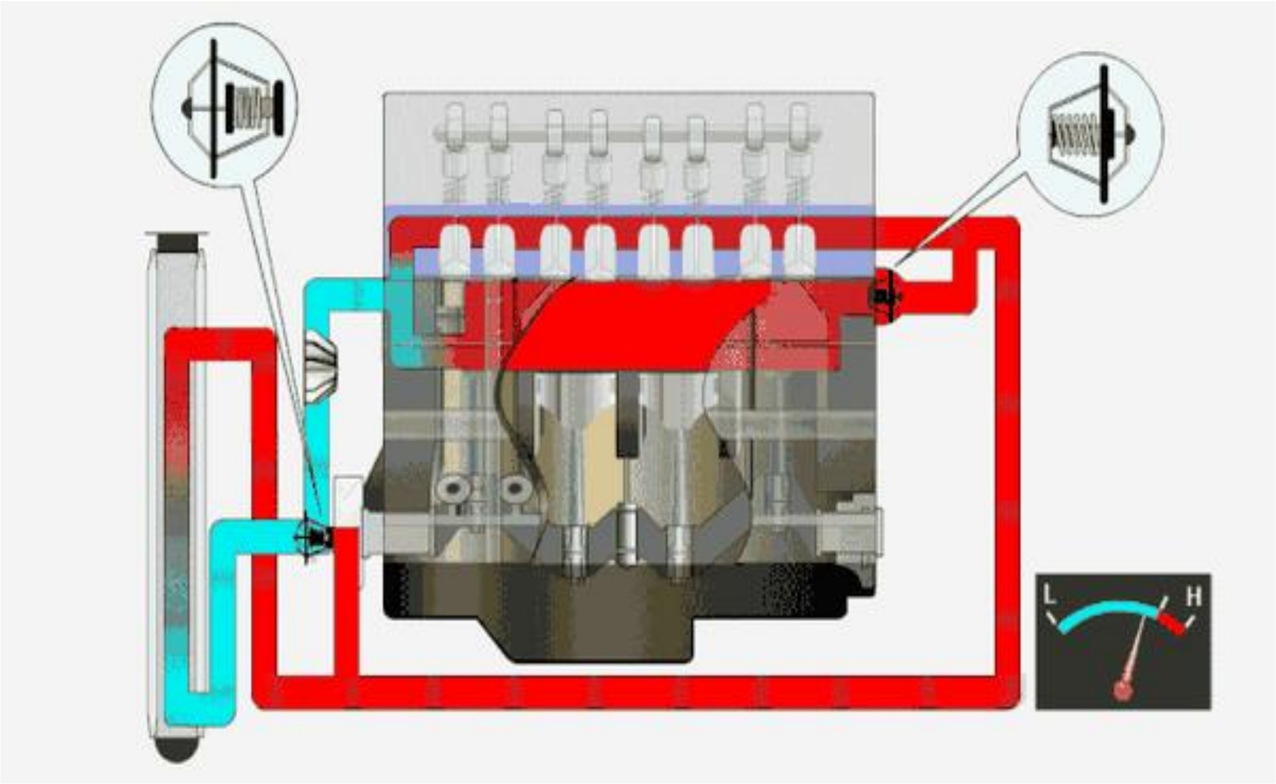
5. Oil jets (piston cooling jets)
– Spray oil under the piston and cool its crown (~700 °C)

6. Pressure relief valve
– Regulates oil pressure and prevents excess pressure

The role of antifreeze

Two temperature levels to hold

– Keep the engine at a safe temperature, ~90 to 100 °C overall
– Cool the walls, ~300 °C locally right after combustion



Antifreeze

The circuit and its seven components

- **1. Coolant (antifreeze)**
  – A mix of water and anticorrosion and antifreeze additives, it absorbs the engine's heat
- **2. Water pump**
  – Circulates the coolant through the system
- **3. Water jackets**
  – Passages around the cylinder and the cylinder head
- **4. Radiator**
  – Cools the coolant with outside air
- **5. Thermostat**
  – Regulates temperature: closed when cold, open when hot
- **6. Fan**
  – Forces air through the radiator
- **7. Expansion tank**
  – Absorbs the expansion of the coolant

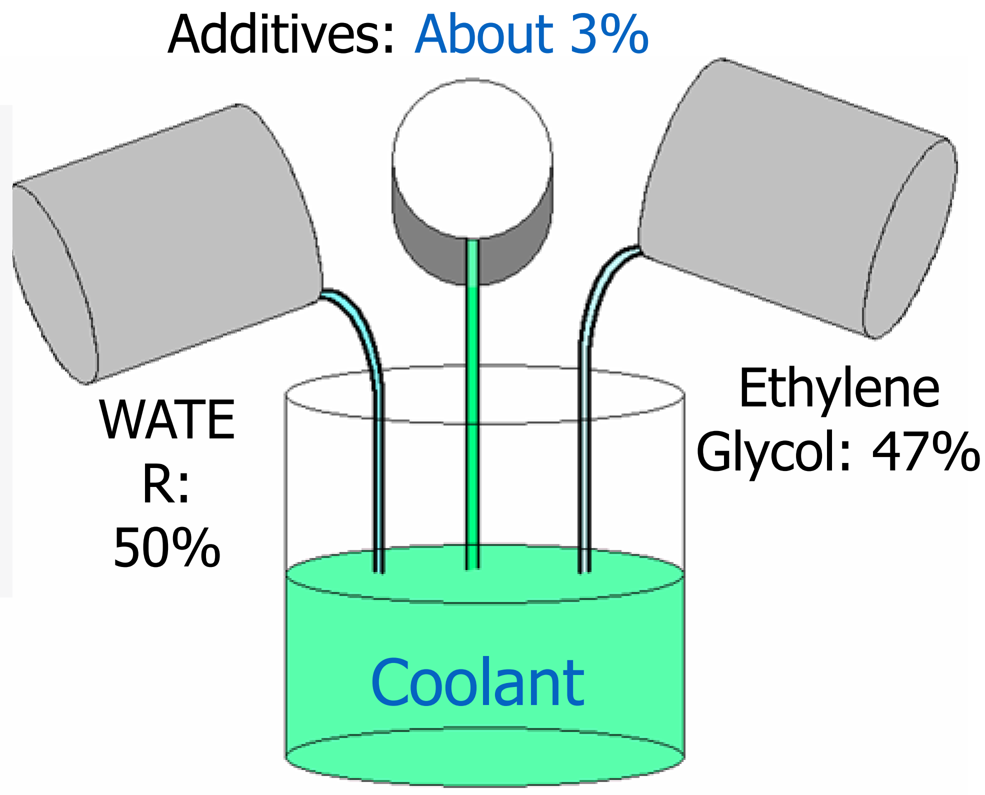
What is coolant made of?

Three ingredients

– Water: 50 %

– Antifreeze, ethylene glycol: about 47 %

– Additives: about 3 %



Why use water?

Thermal property	Water	Glycol (pure)	Water + glycol mix (typically 50–50 %)
Specific heat capacity (kJ	~4.18 (very high)	~2.3–2.5 (lower)	~3.2–3.8 (drops as glycol increases)
Thermal conductivity	~0.6	~0.25	~0.4–0.5
Viscosity	Low (good circulation)	High (resistance to flow)	Medium to high
Freezing point	0 °C	~ -12 to -59 °C (depending on type)	Can go down to ~ -15 to -40 °C
Boiling point	100 °C	>180 °C	105–110 °C
Heat transfer capacity	Excellent	Medium to low	Good, but lower than water
Thermal stability	Good	Very good	Very good
Freeze protection	None	Excellent	Excellent
Corrosion protection	Low (without additives)	May include inhibitors	Good with additives
Cost	Very low	Higher	In between

Coolant requirements

What a coolant must do

– Transfer heat
– Protect against freezing
– Protect against boiling
– Protect against corrosion
– Protect against cavitation
– Protect against foaming
– Stabilize pH

Typical composition, according to Prestone

– Ethylene glycol: 95 %
– Inhibitors, inorganic or organic: 1.5 to 4.0 %
– Water: 1.0 to 3.5 %
– Antifoam agent: 10 to 1,000 ppm
– Dye: trace

ASTM water specifications

Why water

– Inexpensive
– Excellent thermal conductivity
– Very good heat absorption
– Freezes at 0 °C, boils at 100 °C

What the water must meet

– Chlorides < 40 ppm
– Sulfates < 100 ppm
– Calcium < 100 ppm
– Magnesium < 100 ppm
– Total hardness < 170 ppm
– pH from 5.5 to 9.0
– Iron < 1 ppm

North America, Europe and Asia

Water quality varies by region

– In North America, the water is high in chlorides and sulfate ions, while mineral content varies

– In Europe, the water has a very high mineral content

– In Japan, the water is of excellent quality

Coolant additives

Additives are selected according to

– Water quality
– Water pump design
– The metals in the cooling system
– The plastics in the cooling system
– Cylinder liner design
– Drain intervals

KEY POINT — Additives must be selected according to the operating conditions and the conditions listed above.

Ethylene glycol variants

The four families

– Inorganic additive technology: IAT, and heavy-duty IAT
– Organic additive technology: OAT
– OAT with silicates: HOAT
– OAT with phosphates: POAT

KEY POINT — OAT is better for aluminum and has a longer service life.

Type	Full name	Technology	Typical color	Service life	Compatibility	Advantages	Drawbacks	Common use
IAT	Inorganic Acid Technology	Mineral additives (silicates, phosphates)	Green	2 to 3 years (40,000–60,000 km)	Older vehicles	Fast corrosion protection	Short service life, possible deposits	Cars before ~1995
OAT	Organic Acid Technology	Organic acids (carboxylates)	Orange, red	5 years or more (up to 250,000 km)	Modern vehicles	Long life, better thermal stability	Slow initial protection, incompatible with older systems	Modern cars (GM, VW, etc.)
HOAT	Hybrid Organic Acid Technology	OAT + silicates blend	Yellow, turquoise, blue	5 years (~150,000–250,000 km)	Broad compatibility	Good immediate protection + long life	More complex, must match the manufacturer spec	Mercedes, BMW, Ford, Chrysler
POAT	Phosphate Organic Acid Technology	OAT + phosphates (silicate-free)	Pink, light blue	5 years or more	Asian vehicles	Very good aluminum protection, and fast	Brand-specific	Toyota, Honda, Nissan

Coolant specifications

ASTM D 3306, the minimum standard

– Low or high silicate coolants, for passenger vehicles only

ASTM D 4985

– Low silicate coolants for passenger vehicles, and partially formulated coolants for heavy-duty diesel vehicles

– Additives must be added for diesel engines

ASTM D 6210

– Fully formulated coolants for heavy-duty diesel vehicles or passenger vehicles

IAT (green)

Inorganic Additive Technology (IAT) contains

– Silicates
– Phosphates
– Borates

WARNING — IAT is considered an old technology: it can damage ceramic-phenolic seals and the new generation of water pumps.

Silicates

What they bring

Their strengths

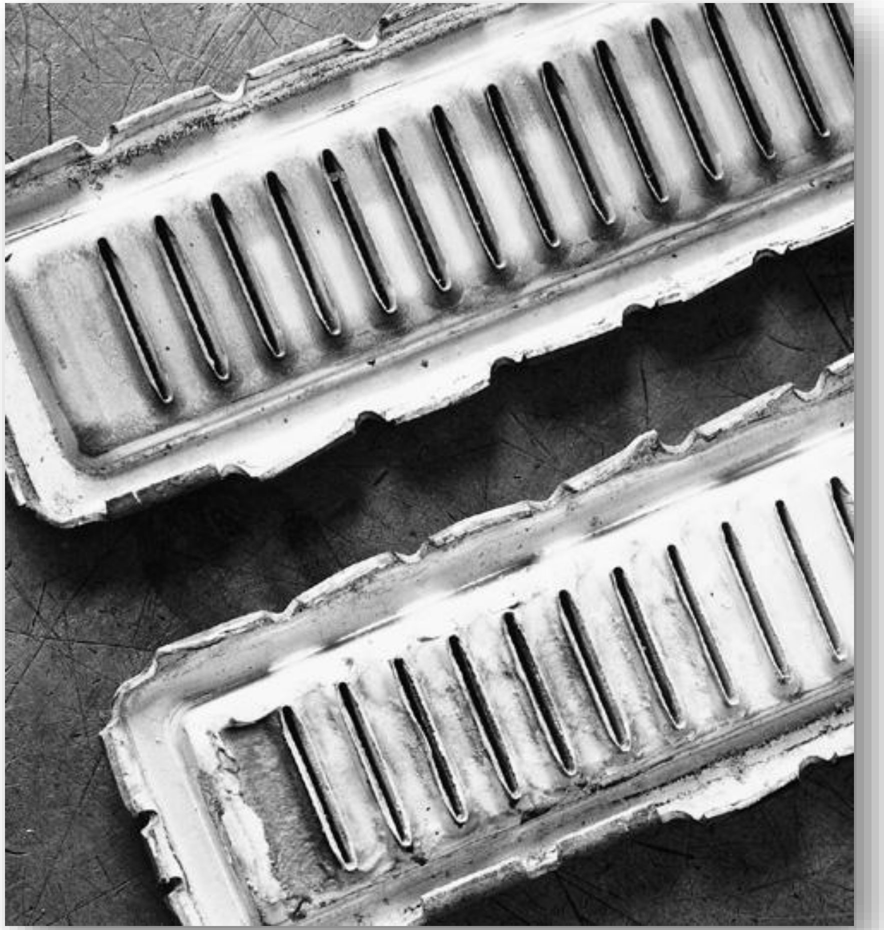
– They offer the fastest protection on the market
– They are the best additives against abrasive contamination and water pump cavitation
– Deposits form very quickly on aluminum surfaces

Silicates

Their downsides

What to watch for

– Deposits tend to form on water pump seals, which promotes seepage
– Silicates eventually drop out of solution and form gels that hinder heat transfer
– They can even break down and become abrasive



KEY POINT — In well-balanced formulations, silicates can last a very long time.

Phosphates

What they do

– Used to prevent corrosion and cavitation of ferrous metals
– They react with the minerals found in hard water, which causes problems
– They are banned by some manufacturers

Borates

What they do

– They provide excellent pH buffering
– In some cases, they can cause aluminum corrosion
– They react poorly when mixed with other additives: silicates, phosphates, etc.

WARNING — Borates are banned by some manufacturers. Some European manufacturers recommend them and others do not, which causes confusion in Europe.

High-silicate coolant

Ordinary green coolants (IAT), high in silicates and phosphates

– High initial pH, from 9.0 to 11.0
– Excellent fast-acting protection against cavitation
– Several quality levels

Their limits

– Short service life: pH drops quickly, change at 8.5
– Silicate deposits can be a problem, the drop in concentration leading to deposits
– Can cause problems with some water pump seals

OAT — the orange family

Organic Acid Technology

What it is

– Organic Acid Technology (OAT), the DEX-COOL® formulation, chosen by GM, VW and several Japanese vehicles
– Designed for long service interval circuits
– FREE of nitrates, amines, phosphates and silicates (NAPS)

Its limits

– More sensitive to dilution and dissolved air
– Can damage plastics under certain conditions
– Long-life protection only in a perfectly maintained circuit

Compatibility

– Not compatible with IAT or with HOAT

OAT Dex-Cool coolant with 2-EHA

What 2-EHA is

– The Dex-Cool additive “sodium 2-EHA” and the other 2-EHA additives are banned by some manufacturers

– They can be a problem with certain plastics

– Many other OAT coolants do not contain any

The real cause of the trouble

– Most problems are caused by contamination

WARNING — Dex-Cool and the other 2-EHA formulations must be considered contaminated even after a minor air intrusion problem, therefore a low coolant level. These systems must be drained, flushed and refilled with a fresh mix, EVEN IF THE COOLANT IN THE SYSTEM IS RECENT.

GM Dex-Cool problems

Low silicate content

Ordinary coolants, low in silicates and high in phosphates

– High pH, from 9.0 to 11.0
– Several quality levels

What can be a problem

– Water quality
– Silicate stabilization
– Quality control

KEY POINT — Some modern low-silicate formulations can last a very long time: 4 years or 100,000 km.

HOAT (several colors)

Hybrid Organic Acid Technology

– Found in newer Ford, Chrysler and Mercedes vehicles

– It combines the best qualities of IAT and OAT: it protects very well and lasts a very long time

Nitrites

What they do

– They provide excellent protection against cylinder liner cavitation and help control corrosion
– They deplete in service and must be recharged, in the form of heavy-duty additives

WARNING — Too high a nitrite concentration can become a problem. Nitrites are carcinogenic and are banned in certain cases.

HOAT (G-05)

What hybrid coolants are

– They pair organic and inorganic (silicate) additives
– Used for factory fill by European manufacturers as well as by Ford and Chrysler

Their characteristics

– Low pH, 7.5
– Phosphate-free
– Excellent protection against cavitation
– Partial and fully formulated versions available

KEY POINT — It combines the best of both worlds.

POAT Coolant

Phosphated Organic Acid Technology (POAT)

– Dark green
– Used in Mazda and Ford 2008 and newer
– Mazda standard FL-22

Coolant pH

The lowest pH coolants

– BMW's G48 and G05 coolants are among the lowest pH coolants, therefore the most acidic

In the shop

– Test specifications must be adjusted to the actual results

– Some evidence suggests that BMW's G48 may be acceptable below a pH of 7

KEY POINT — Change thresholds: IAT at 8.5; recent OAT and HOAT at 7.5.

Coolant and pH

Factory fill coolants, Asian and European manufacturers and GM

– Low pH, from 7.8 to 8.3, silicate-free and phosphate-free
– OAT with silicate = HOAT, OAT with phosphate = POAT

What can be a problem

– Protection takes time before it becomes effective
– Cavitation
– Plastics
– Quality control
– 2-EHA additives

KEY POINT — Traditional IAT coolants started at a pH of 10.5 to 11.0 but dropped faster. The lower pH of OAT is preferable with aluminum: an OAT does not need replacing until it falls below 7.5.

“G” designations

What the G stands for

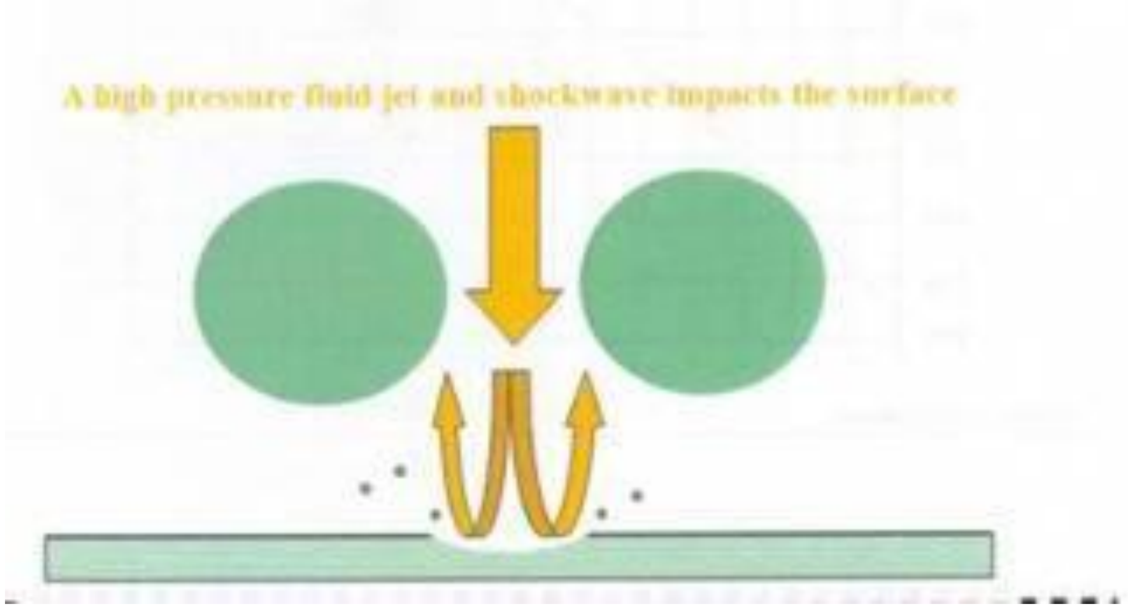
– G = Glysantin, Valvoline (Zerex)

The designations

– G11: VW before 1997, blue
– G12: VW 1997 and newer, red or pink
– G12+: VW 2003 and newer, phosphate-free HOAT, violet
– G40: replaces G12+
– G30, G33, G34: silicate-free and phosphate-free, Dex-Cool replacement
– G05: phosphate-free, very low silicate content, formulation for Japanese vehicles and Chrysler HOAT
– G48: very low silicate and phosphate content, blue
– NAP: free of nitrates, amines and phosphates (BMW)

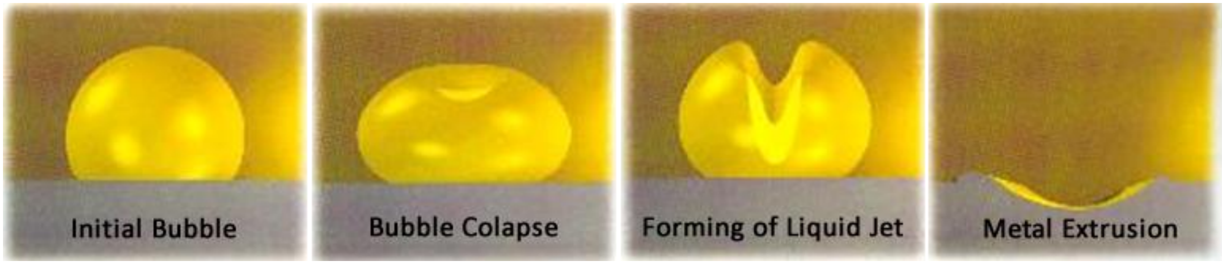
Cavitation

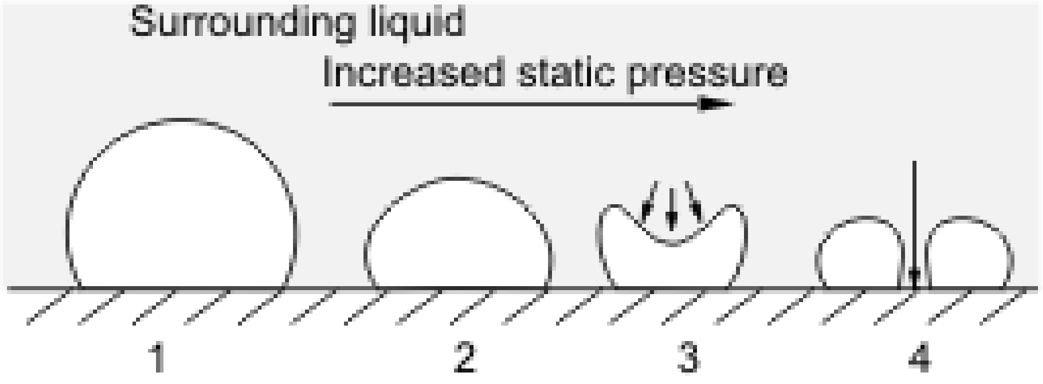
What happens in the coolant



The mechanism

– When a vapor bubble implodes against a rigid surface, it damages that surface
– Cavitation produces a continuous stream of bubbles that damage the surface

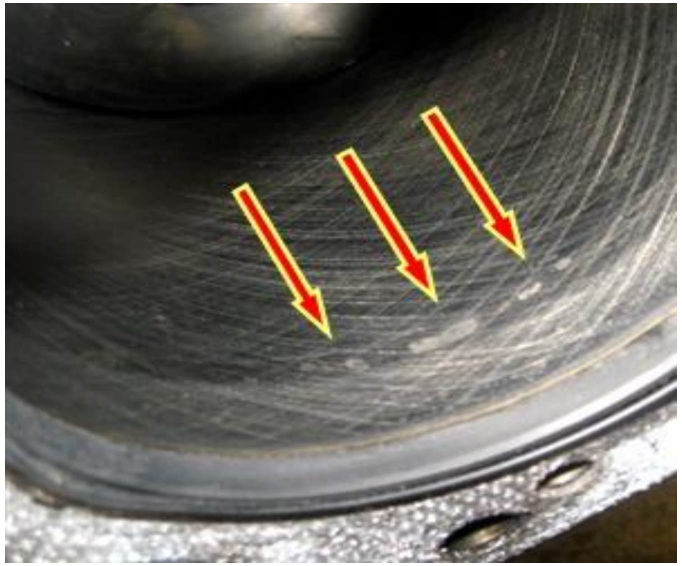




Cavitation bubble imploding close to a fixed surface generating a jet (4) of the surrounding liquid.

Cavitation

Where it strikes



The areas affected

- It has always been a serious problem in diesel engine cylinder liners
- It also causes problems in gasoline engine water pumps
- Aluminum is far more sensitive to it than other materials
- Some engines are far more prone to it than others

Cavitation

What it can destroy

How fast the damage happens

- Cavitation can destroy a water pump in as little as 60 days
- A diluted coolant or the wrong coolant can be the cause
- Any restriction that increases water pump suction will also increase cavitation
- Some pumps are more prone to it than others

Cavitation, the first sign of contamination

What to look at

– Coolant type, dilution, etc.
– Radiator cap
– Modifications
– Minor restrictions
– Extreme conditions
– Budget components

Cavitation, the first sign of contamination

Plastic impeller here

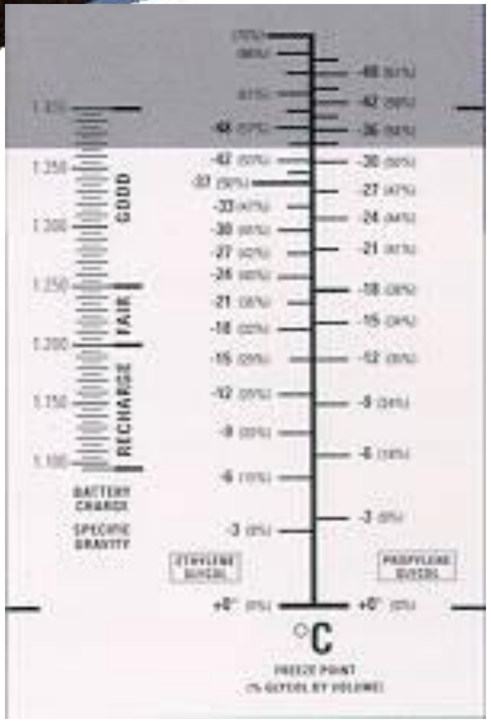
A growing problem

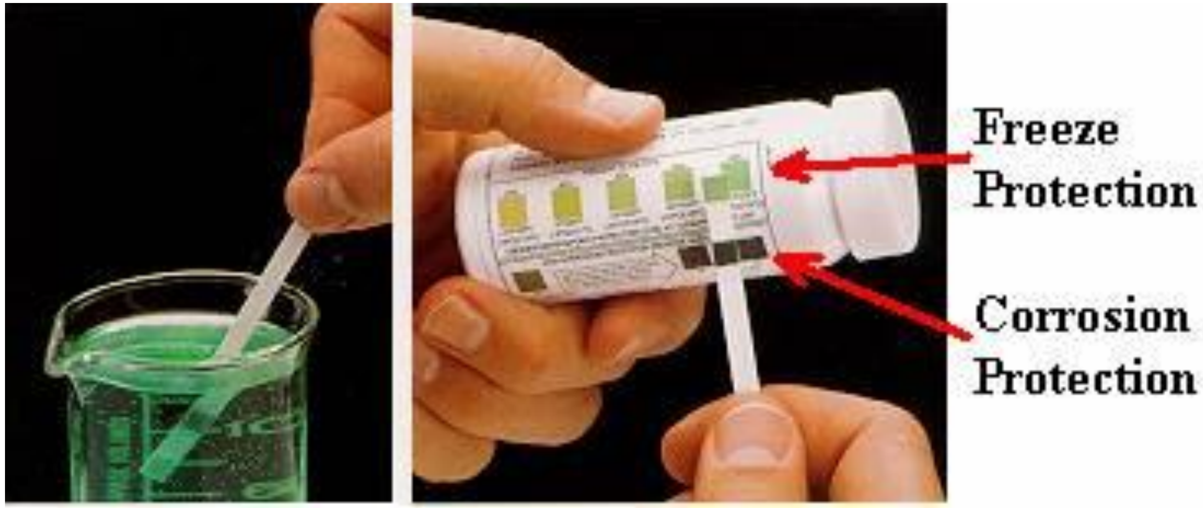
– Cavitation damage is becoming more and more common
– Many cars require a coolant offering extra protection against cavitation

Key point

– There is no such thing as a universal coolant

Density test





pH tests


RadStrips


EXTECH


EXTECH
INSTRUMENTS
14.0
79.1 °F
pH
pH/ORP/Temperature Meter
PH210
/ESC
FUNCTION
HOLD
CAL
REC
ENTER
SETTING

Careful: electrolysis

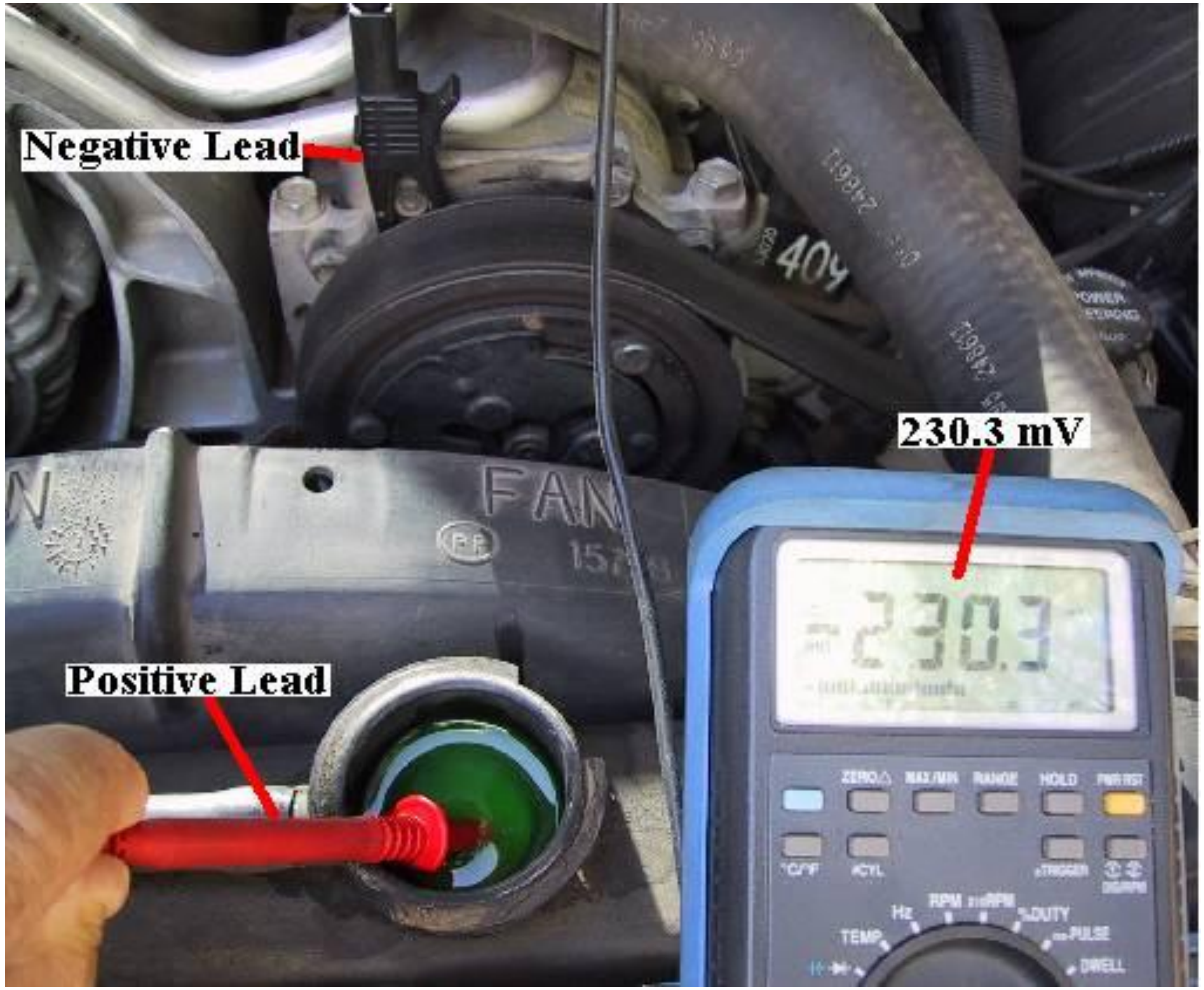
What gets wrongly blamed on it

– Many technicians assume that any unexplained damage is caused by electrolysis
– It is often blamed whenever there is cavitation or erosion-corrosion of the water pump
– These problems are in fact often caused by aluminum oxide contamination

Real electrolysis

– Poor grounds can send the electron flow from the starter or the alternator through the cooling system
– This can damage a heat exchanger in a week or less
– Voltage drop tests should be performed on the ground circuits

Coolant voltage test



How to measure

– Coolant voltage still has to be measured
– All loads must be switched on
– The acceptable limit is about 0.3 V

Plastic component failure

Plastic has taken the place of metal

– It has replaced aluminum, steel, copper and brass in some components, including some seals
– The most widely used plastic is nylon 6.6, whose critical temperature is high

WARNING — Nylon 6.6 is also sensitive to some of the organic acids (OAT) used in certain coolants.

Thermostats

What a car thermostat does

What it does

– It regulates coolant temperature and controls its flow to the radiator

Temperature sensing

– The thermostat senses the coolant temperature and reacts by opening or closing the valve

Warm-up phase

– As the engine warms up, the coolant temperature rises and causes the valve to open

Coolant circulation

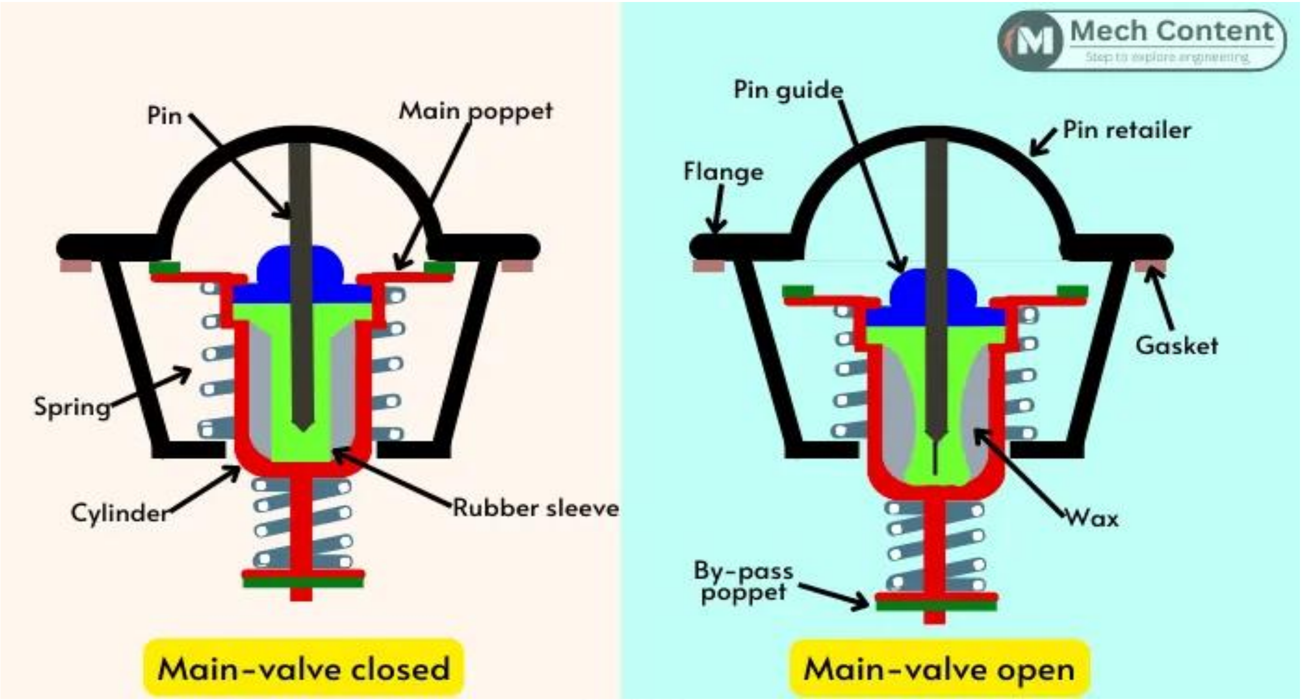
– The coolant then flows to the radiator to release its heat

WARNING — A faulty thermostat leads to engine overheating, excessive fuel consumption and increased pollutant emissions. Hence the importance of checking it and replacing it when needed.

Wax thermostat

The thermostatic element

- **1. Wax thermostatic element (wax motor)**
  – The main component of the thermostat, responsible for opening and closing the valves according to coolant temperature
- **1a. Brass or copper cylinder**
  – Metal housing filled with wax, in contact with the coolant, it transfers heat to the wax
- **1b. Rubber sleeve**
  – Seals the wax in; as the wax expands, it compresses and pushes the pin outward
- **1c. Piston or pin**
  – Fitted in the sleeve, it moves as the wax expands to actuate the valves
- **1d. Piston guide**
  – Holds and guides the piston travel, for precise and stable movement



Wax thermostat

Valves and retainer

2. Main valve

– Controls coolant flow to the radiator
– Linked to the wax cylinder, it opens progressively as temperature rises

3. Pin retainer

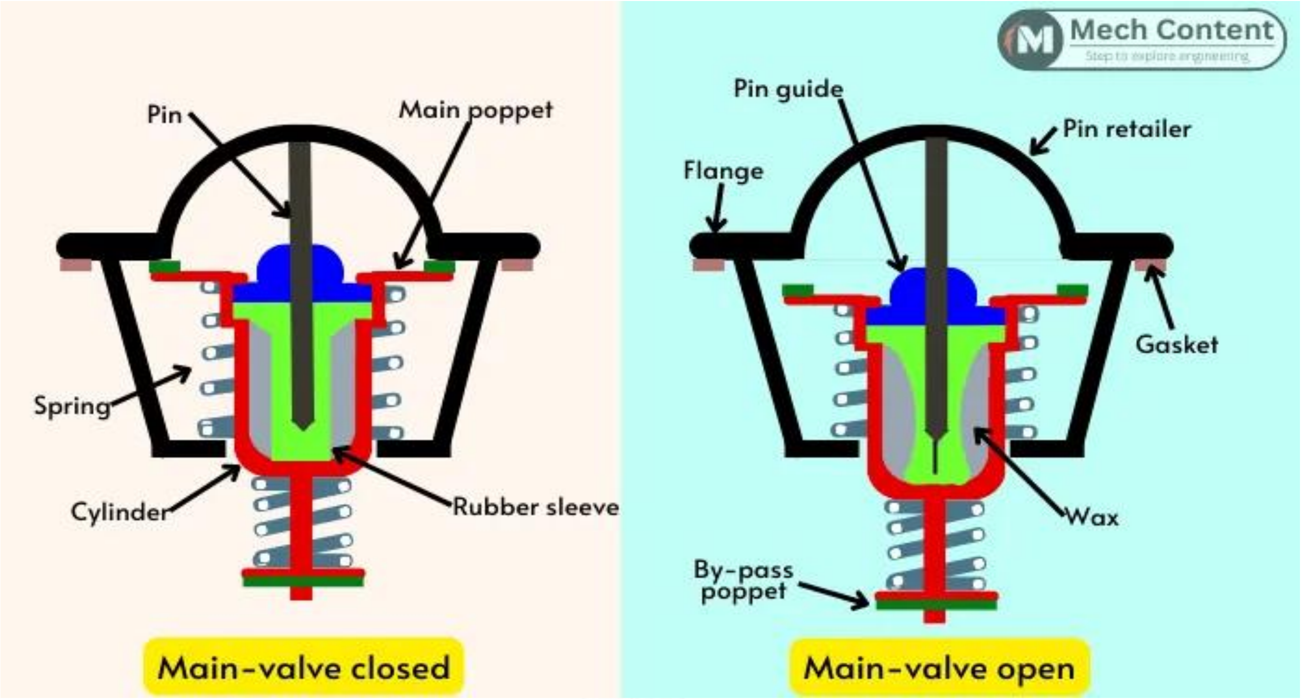
– Holds the end of the piston in place; as the wax expands, it lets the cylinder move while keeping the pin stable

4. By-pass valve

– Controls the flow of coolant through the by-pass circuit

Key point

– Engine at normal temperature: the main valve opens to let the coolant cool down in the radiator
– Cold engine: the coolant flows mainly through the by-pass circuit, for a fast warm-up



Wax thermostat

Sealing, return and bleeding

5. Seal

– Fitted on the thermostat flange, it prevents leaks and ensures the coolant flows only through the intended passages

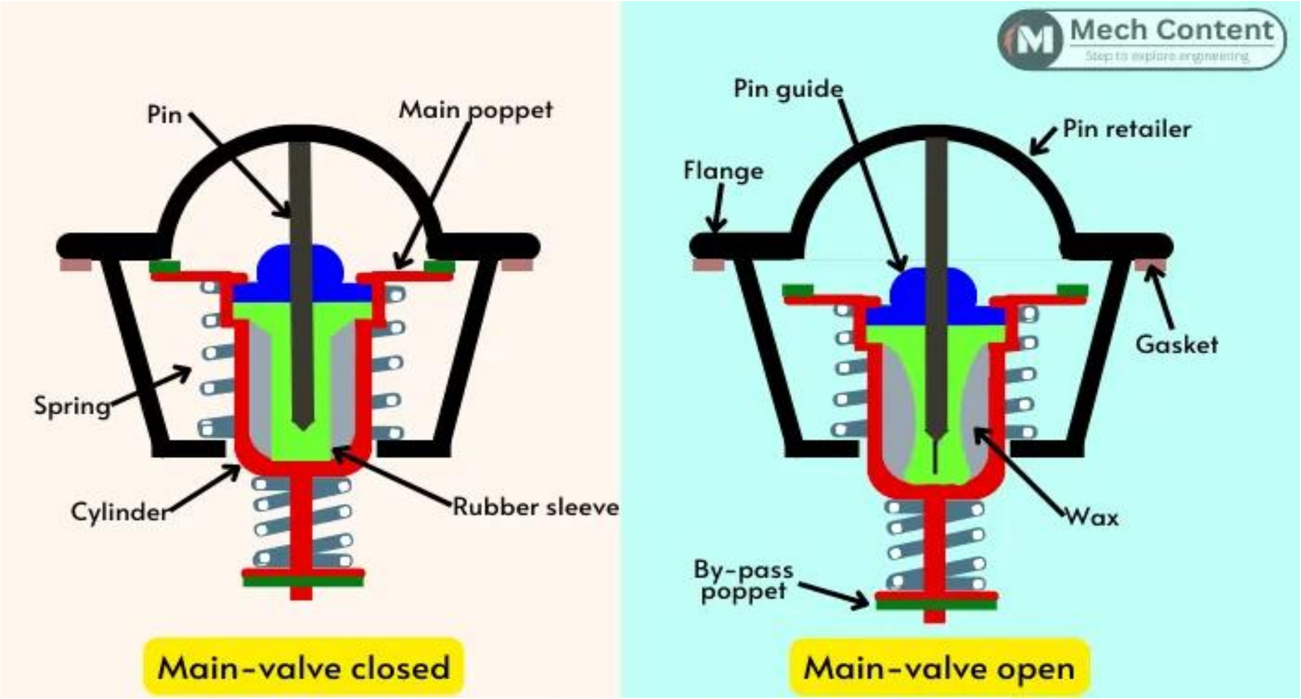
6. Return spring

– Normally holds the main valve closed; as temperature drops, it returns the thermostat to its initial position

7. Bleed pin (jiggle pin)

– Vents air trapped in the circuit to the radiator, both during filling and in operation

– Helps prevent air pockets, which improves system efficiency



Electronic thermostat

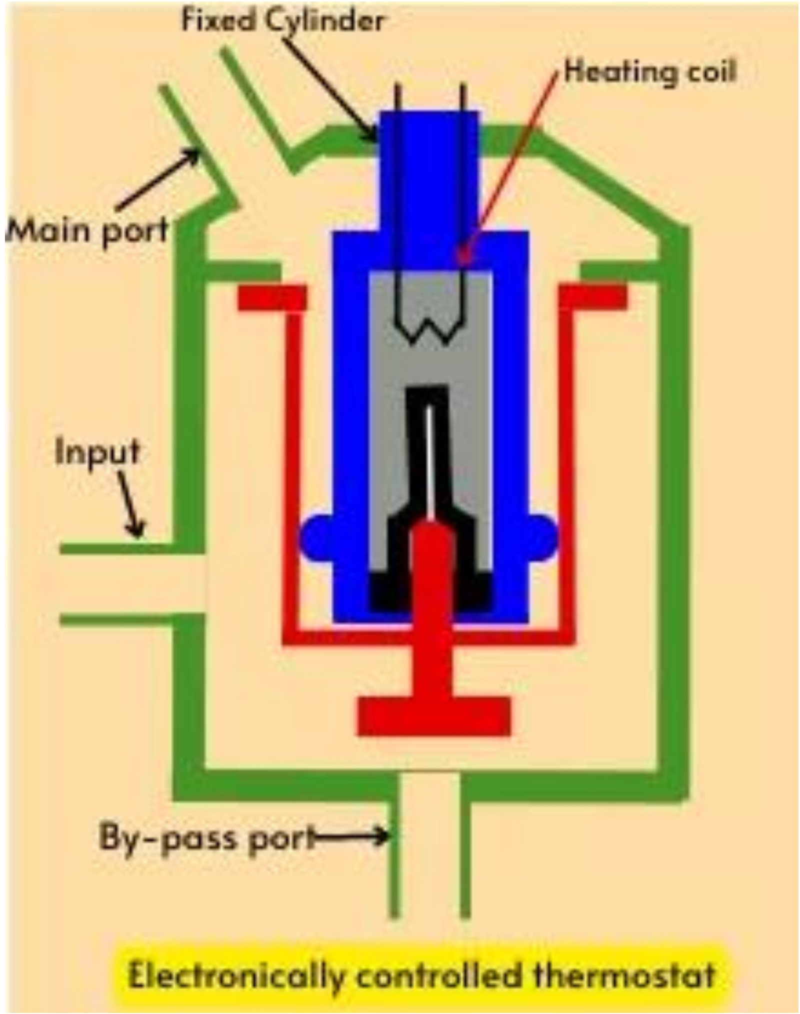
Electrically controlled thermostat

How it is built

– It uses a built-in heating element to heat the wax of the thermostatic element
– The heating element is fitted directly inside the housing holding the wax

What controls it

– Power to this heating element is controlled by the engine control unit (ECU)
– Thermostat opening therefore no longer depends on coolant temperature alone



Electronic thermostat

Operation

With a conventional wax thermostat

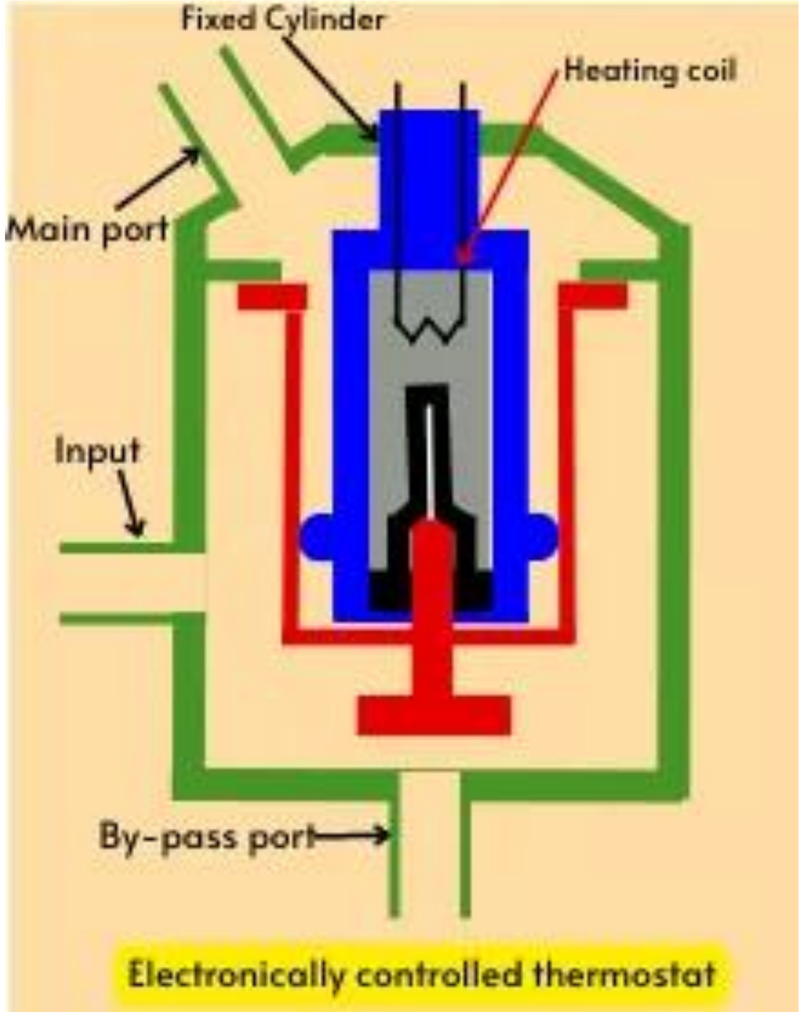
– A rise in engine load raises engine temperature, then coolant temperature
– Once the wax is hot enough, it expands and opens the thermostat
– There is therefore a delay before cooling begins

With an electrically controlled thermostat

– The ECU detects the rise in load and powers the heating element immediately
– The wax expands faster, the thermostat opens early and flow to the radiator increases

Result

– Cooling begins before the coolant temperature has risen significantly



Electronic thermostat

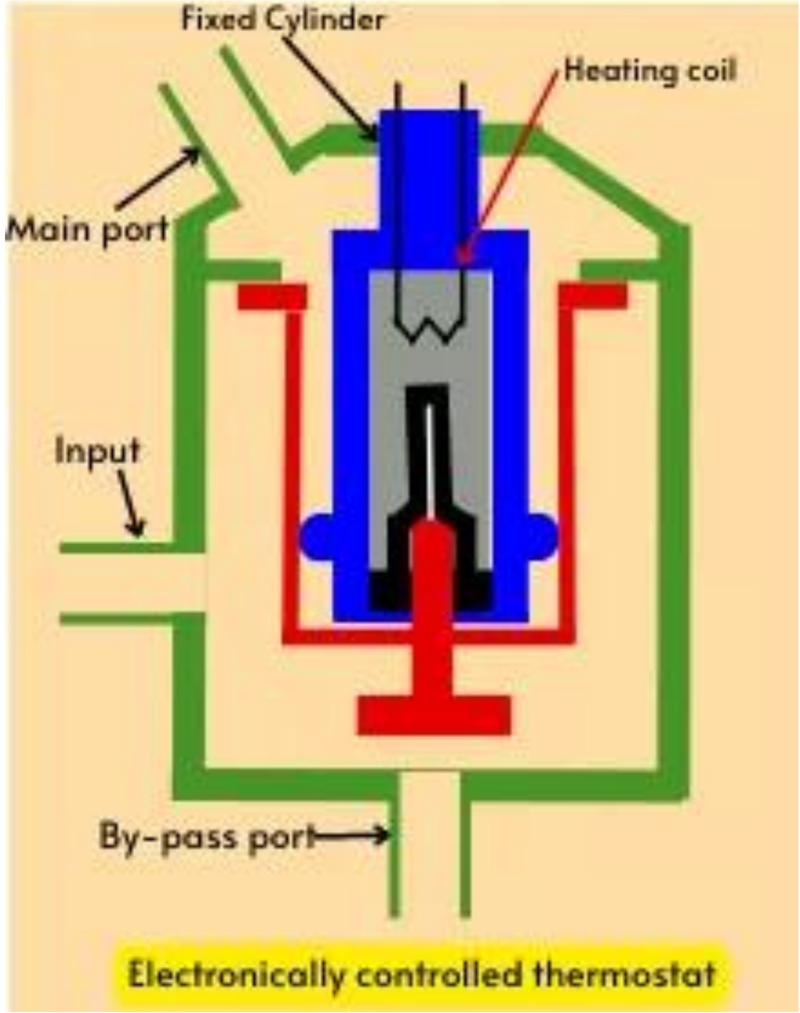
Advantages

What you gain

– Faster reaction than a conventional wax thermostat
– Better control of engine temperature
– Operation matched to engine load and speed
– Improved engine efficiency and thermal management

Key point

– The electrically controlled thermostat lets the ECU control opening according to engine operating conditions, rather than waiting on coolant temperature alone



The water pump

What it does

– It circulates the coolant between the radiator and the engine
– The impeller, driven in rotation, creates a centrifugal force that pushes the coolant through the circuit

The coolant's path

– It flows through the engine block and the cylinder head to absorb the heat produced by combustion
– It returns to the radiator, where that heat is released thanks to the radiator and the fan

KEY POINT — This continuous cycle keeps the engine at its optimal operating temperature, improves its performance and prevents damage caused by overheating.

Types of water pump

– Mechanical
– Variable
– Electric
– Auxiliary

The mechanical water pump

What drives it

– Depending on its design, it can be bolted to the engine or built into the engine block
– Unlike an electric pump, it is driven mechanically by a timing belt, an accessory belt (V-belt) or directly by the engine

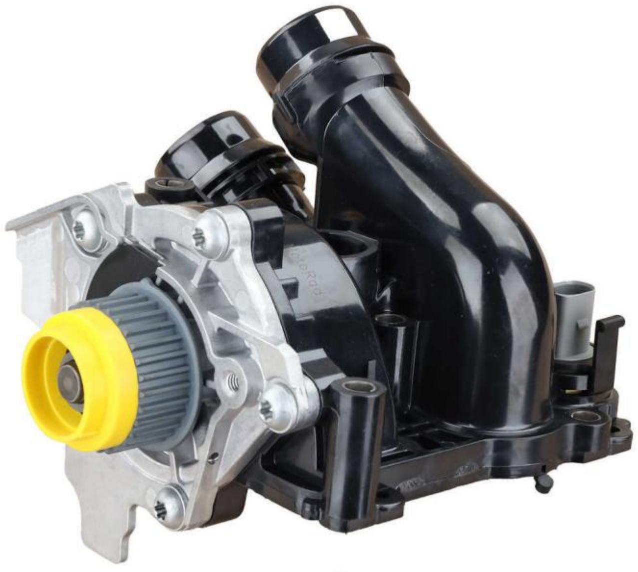
The variable water pump

What it does

– It adjusts its flow according to engine demand
– A control system lets it run only when cooling is needed

What you gain

– Better energy efficiency, reduced fuel consumption and lower pollutant emissions



The electric water pump

What it does

– It produces a coolant flow independent of engine speed

– Its operation is electronically controlled and is not tied to engine speed

What you gain

– Optimized cooling, with reduced power losses, fuel consumption, mechanical friction and emissions

The auxiliary water pump

What it does

– It backs up the main pump by keeping coolant flowing through certain parts of the vehicle
– It is usually fitted on a branch of the main cooling circuit

Where you find it

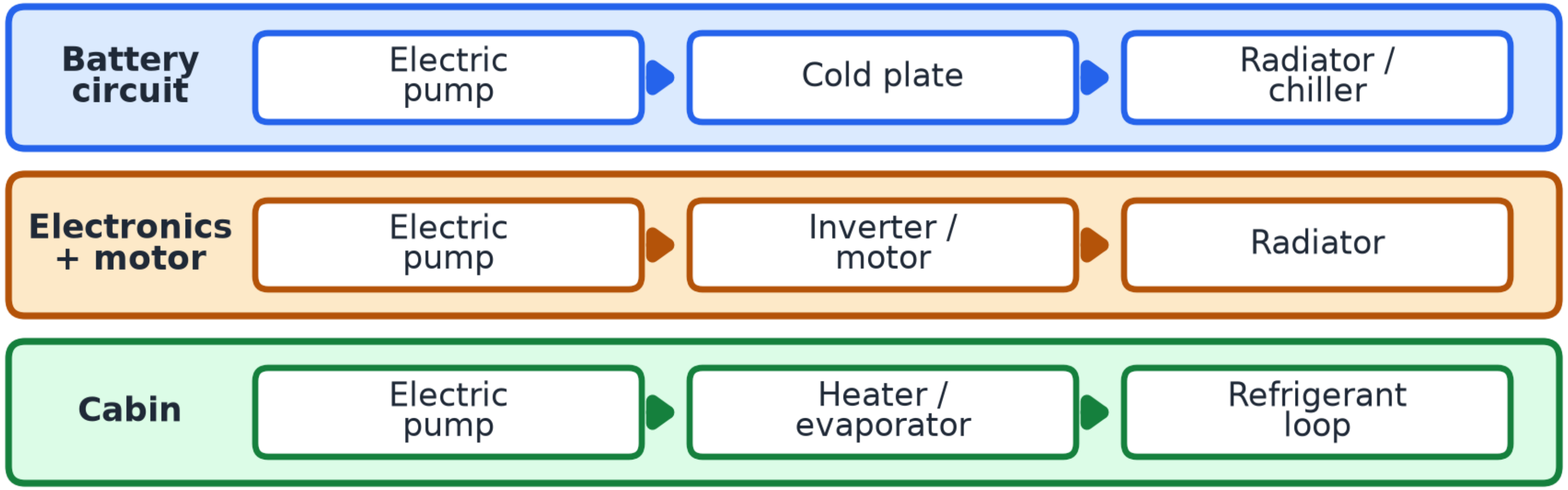
– Mainly on hybrid and electric vehicles, which often have several cooling circuits
– It routes coolant to the high-voltage battery, the power electronics or the electric motor

Cooling an EV

Several circuits, several pumps

How it is organized

– An EV has several separate glycol loops, each with its own electric pump or pumps
– The glycol carries and redistributes the heat; valves link the circuits together



Valves link the circuits: series or parallel, and heat sharing.

Auxiliary pumps

The heart of circulation on an EV

Why they are central

– With no engine belt, all circulation depends on electric pumps; an EV has several, one per circuit plus dedicated pumps

– Controlled by the thermal management module, they match flow to demand and run even with the vehicle parked, while charging

Diagnosis

– A failing pump causes localized overheating of a component, battery or inverter, without the classic engine overheating

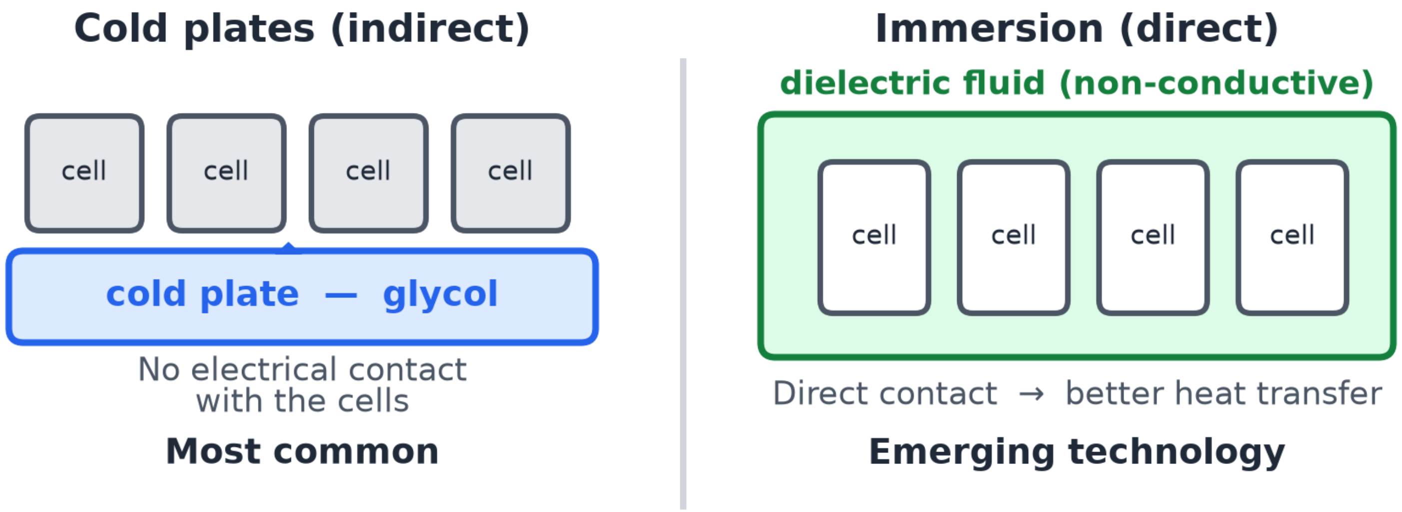
– Look for coolant flow codes (e.g. P00B7) and check the pump's power supply and signal

Cooling the battery

Two main methods

The two approaches

– Cold plates: the glycol flows under the cells, with no electrical contact. The most common
– Immersion: cells in a dielectric fluid. Better transfer, emerging technology



EV-specific coolant

A coolant unlike the others

Low conductivity required

– In a high-voltage environment, the coolant must have low electrical conductivity to prevent leakage currents and corrosion

Recognizing the product

– Often violet in color, meeting a G13-type specification

Shop rules

– Never substitute a conventional combustion-engine antifreeze

– Always use the manufacturer's exact specification (compatible materials and conductivity)

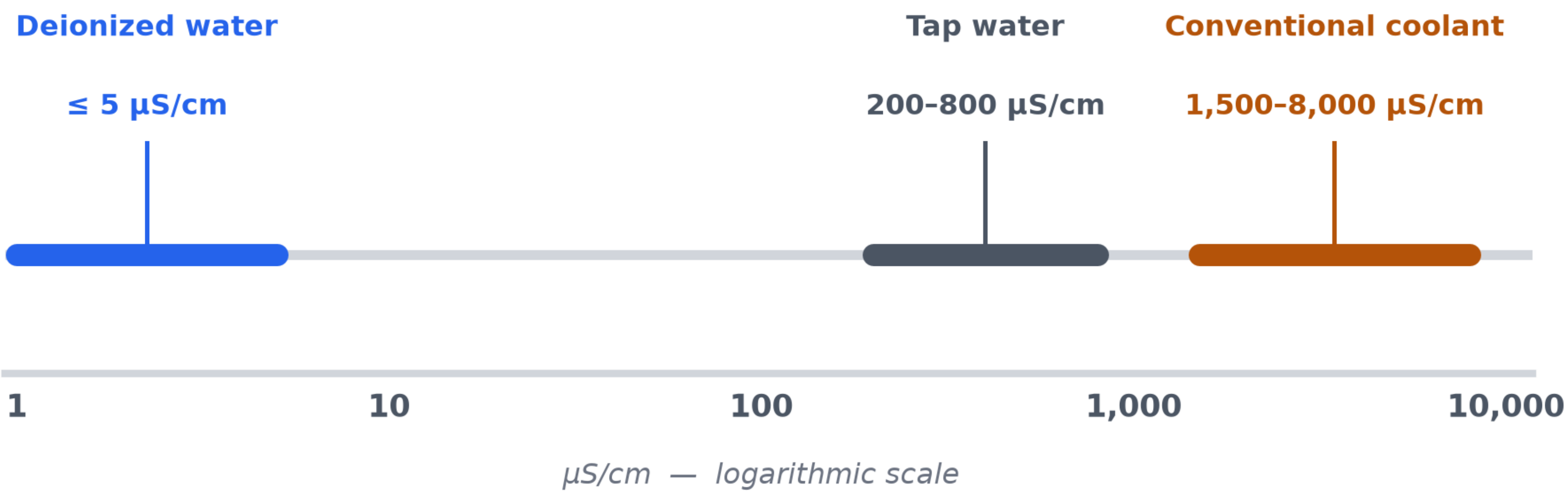
KEY POINT — The antifreeze chemistry (IAT, OAT, HOAT, POAT) covered earlier applies here too, with this added conductivity requirement.

Conductivity (µS/cm)

What it measures

The measurement

– A fluid's ability to conduct current, in µS/cm; it depends on the dissolved ions
– On an EV, a coolant that conducts too well means leakage current, corrosion and fire risk

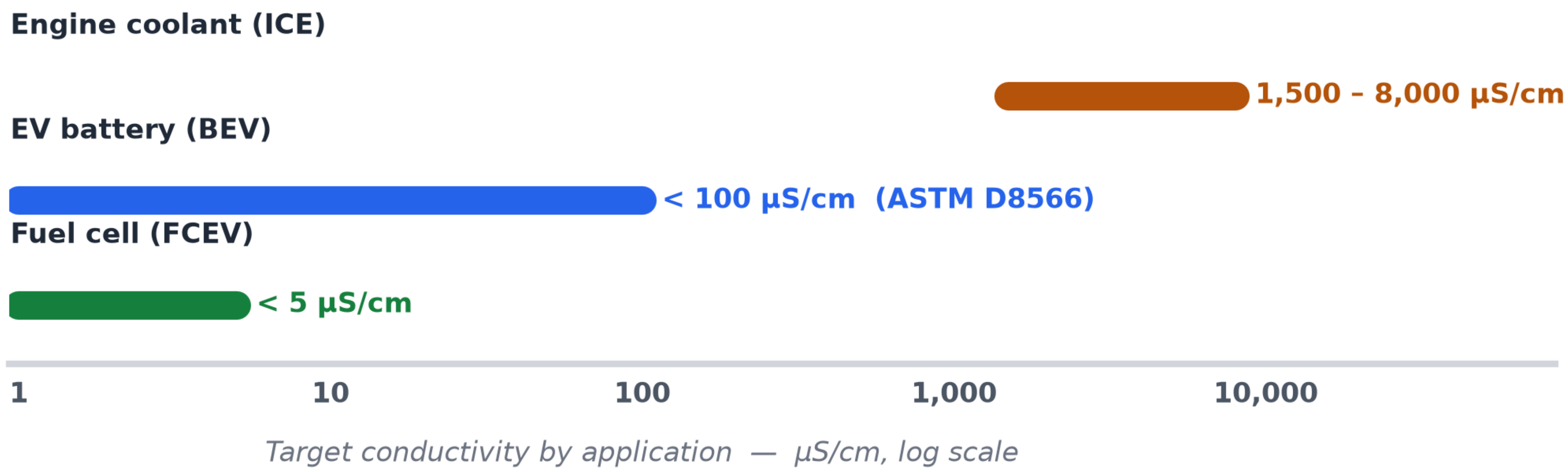


Conductivity values

Service and diagnosis

What you do in the shop

– Deionizer (ion exchange resin): for now only on FCEVs, for example the Toyota Mirai and the Hyundai Nexo
– On battery EVs: measure with a conductivity meter and replace the coolant above the manufacturer's limit



Servicing the EV circuit

Procedures to follow to the letter

The rules

– Air bleeding is almost always done with a scan tool that runs the pumps; it cannot be done manually
– Never mix the circuits or the coolants
– After replacing a component: calibration or reprogramming is often required
– Always follow the manufacturer's procedure, as with any pressurized system

WARNING — High-voltage safety: apply the full HV shutdown procedure covered in the Heat Pump training before any work.

Active thermal management

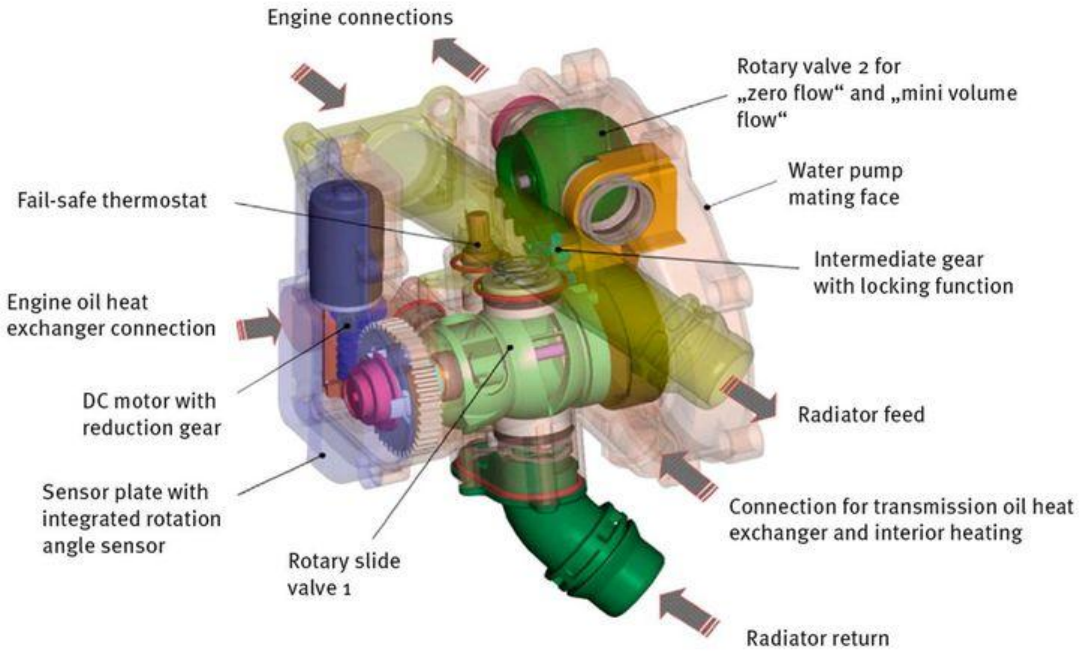
Active thermal management

Why active thermal management?

What it brings

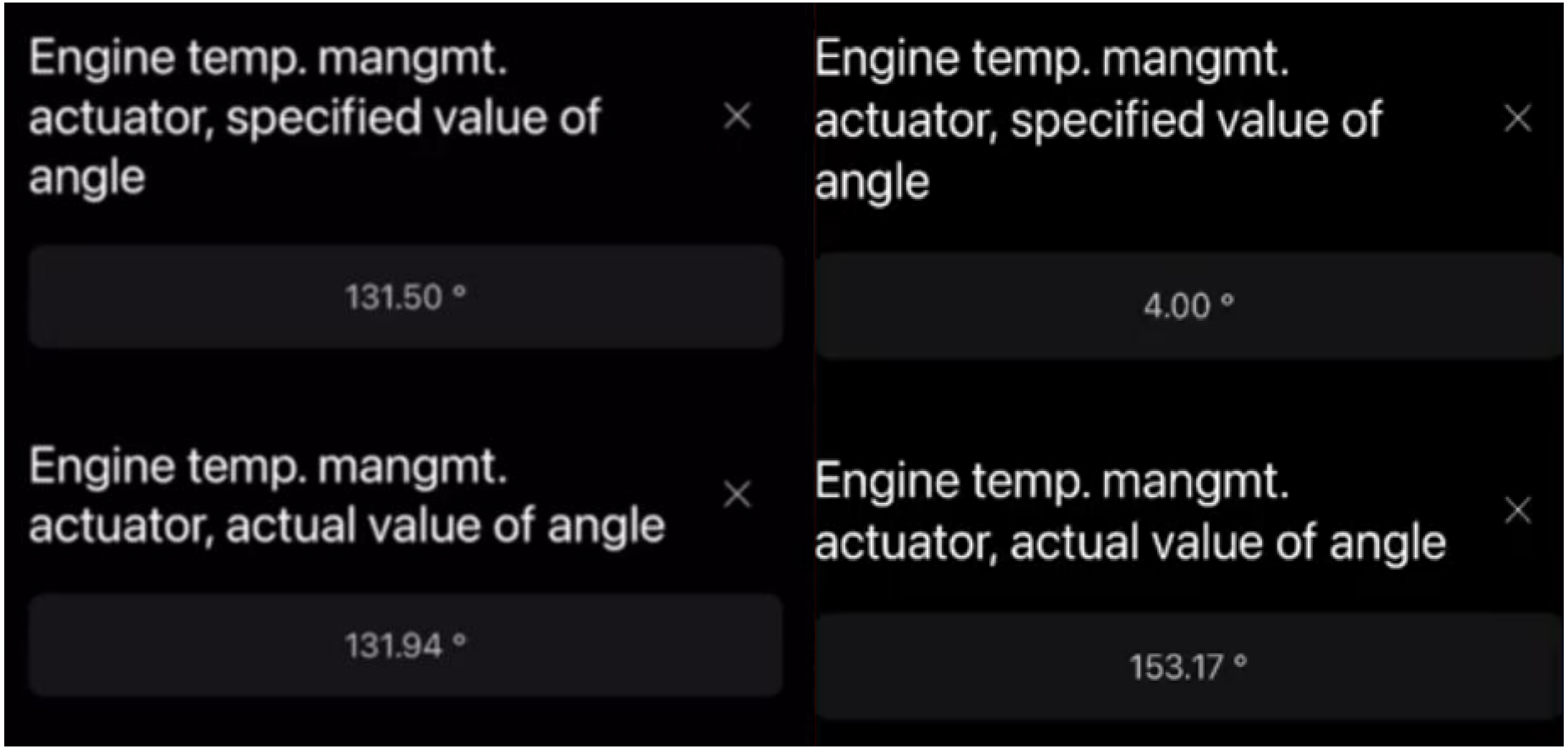
– Speeds up engine warm-up, to reduce wear and improve efficiency
– Reduces fuel consumption by precisely controlling coolant flow
– Lowers pollutant emissions and helps meet environmental standards
– Gives better control of engine, turbocharger and transmission temperature
– Optimizes cooling of batteries and electronics on hybrid and electric vehicles

Audi, variable mechanical pump



Coolant leak through the thermostat's electric motor

P00B700: engine coolant flow low performance



GM, engine coolant flow control valve

Low temperature
(diesel)

High temperature

High temperature

What active thermal management aims for

– Unlike conventional systems with a mechanical pump, it aims above all to optimize fuel economy by keeping the engine at its ideal temperature, whatever the operating conditions and engine speed

– It also delivers the performance the driver expects, while reducing pollutant emissions

A secondary cooling circuit

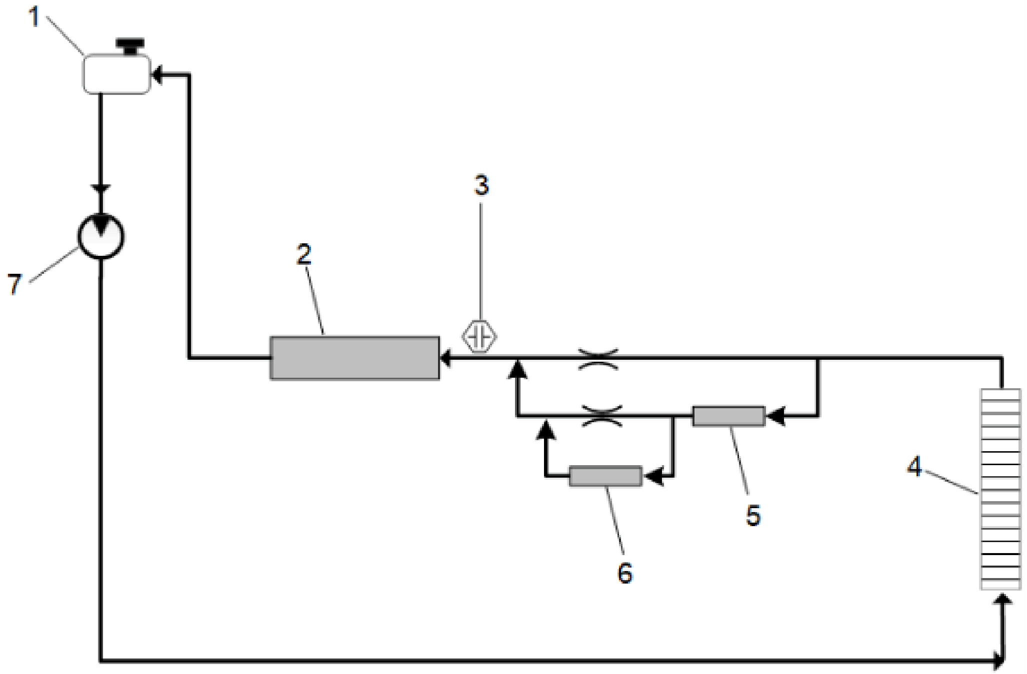
– It feeds the charge air cooler (air-to-water heat exchanger), the fuel cooler and the diesel exhaust fluid (DEF) injector

– Flow is provided by a low-temperature auxiliary radiator and an electric pump mounted on the chassis

Low temperature

The secondary circuit

– It circulates coolant to the water charge air cooler), the fuel cooler and the diesel exhaust fluid (DEF) injector
– It uses an auxiliary engine radiator and an auxiliary coolant pump



Engine coolant flow control valve

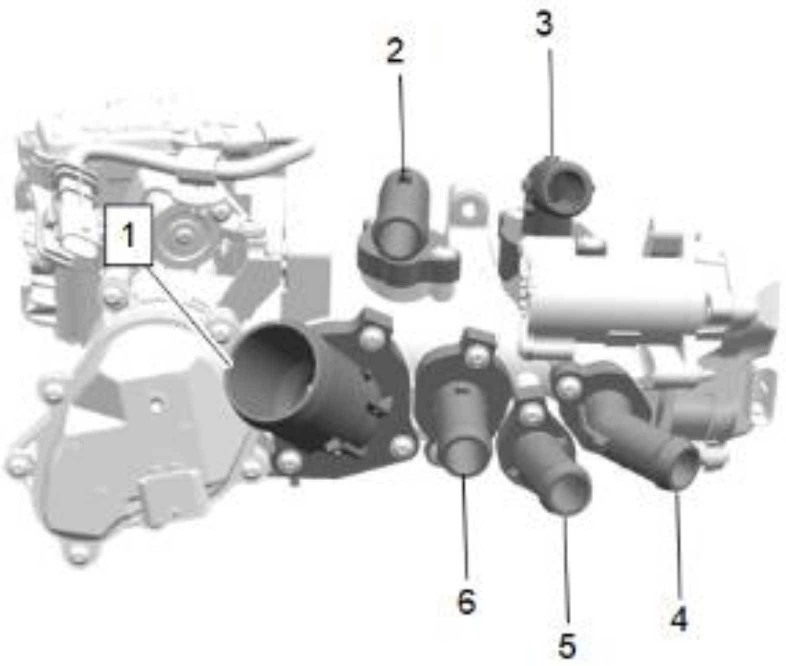
The flow control valve

What it does

– It regulates coolant flow to the radiator as well as to the engine oil and transmission oil heat exchangers
– It replaces the traditional wax thermostat and is electronically controlled by the ECM/ECU from the information supplied by various sensors

Its six ports

– 1. Large side port: controls coolant flow to the radiator
– 2. Upper port, always open: feeds the EGR system, the turbocharger and the cabin heater core
– 3. Right side port: allows coolant returning from the EGR and the turbocharger
– 4. Lower right port: receives the cooled coolant coming from the heater circuit auxiliary pump
– 5. Small valve to the left of port 4: controls flow to the engine oil and transmission oil heat exchangers
– 6. By-pass port: keeps coolant flowing through the by-pass circuit when flow to the radiator is not required

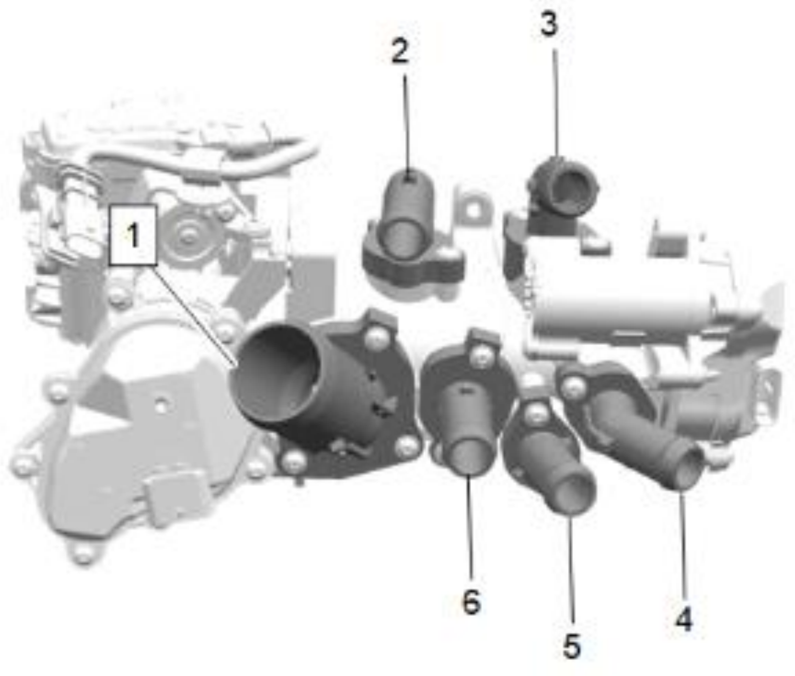


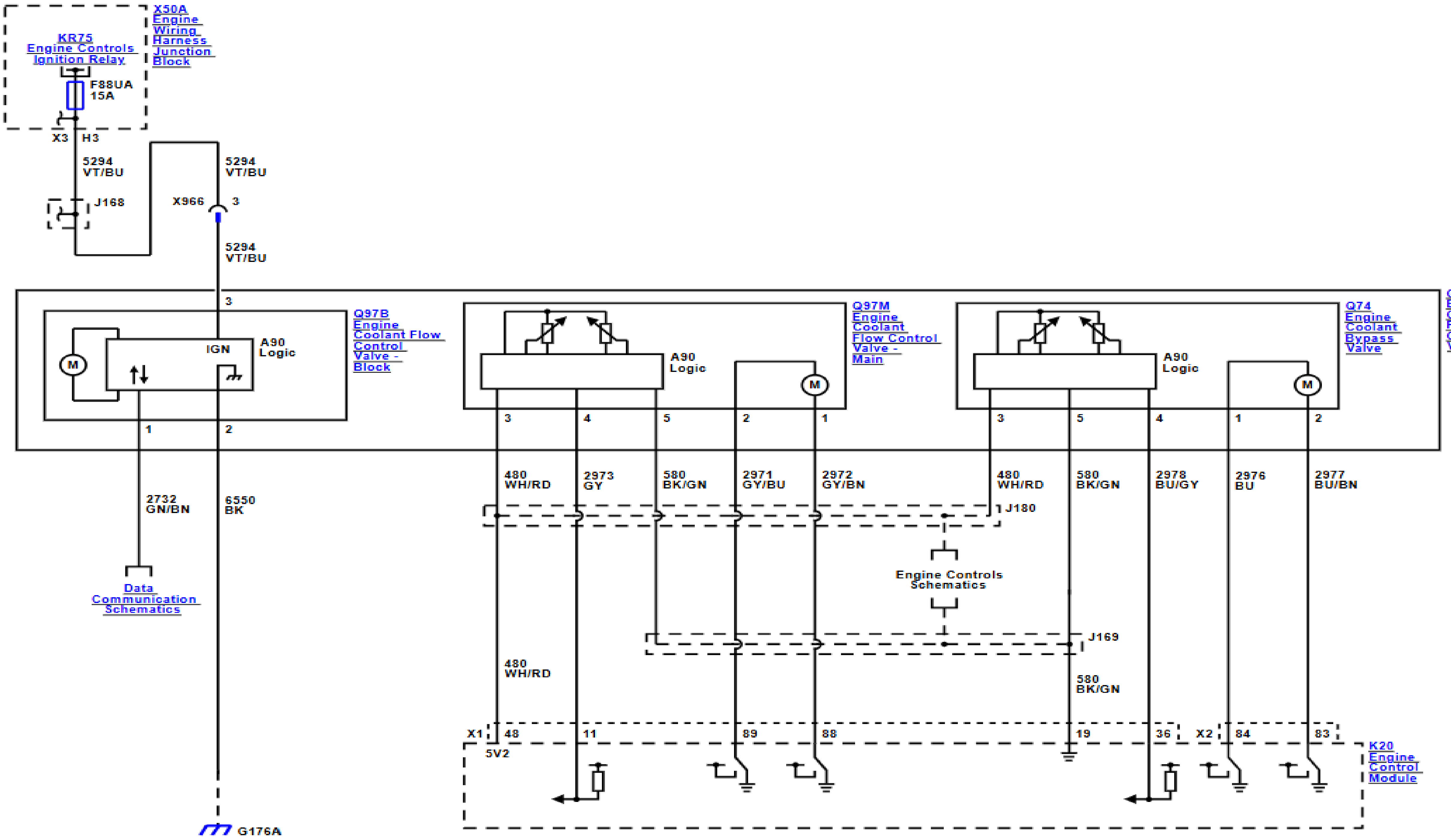
Engine coolant flow control valve

Seven operating modes

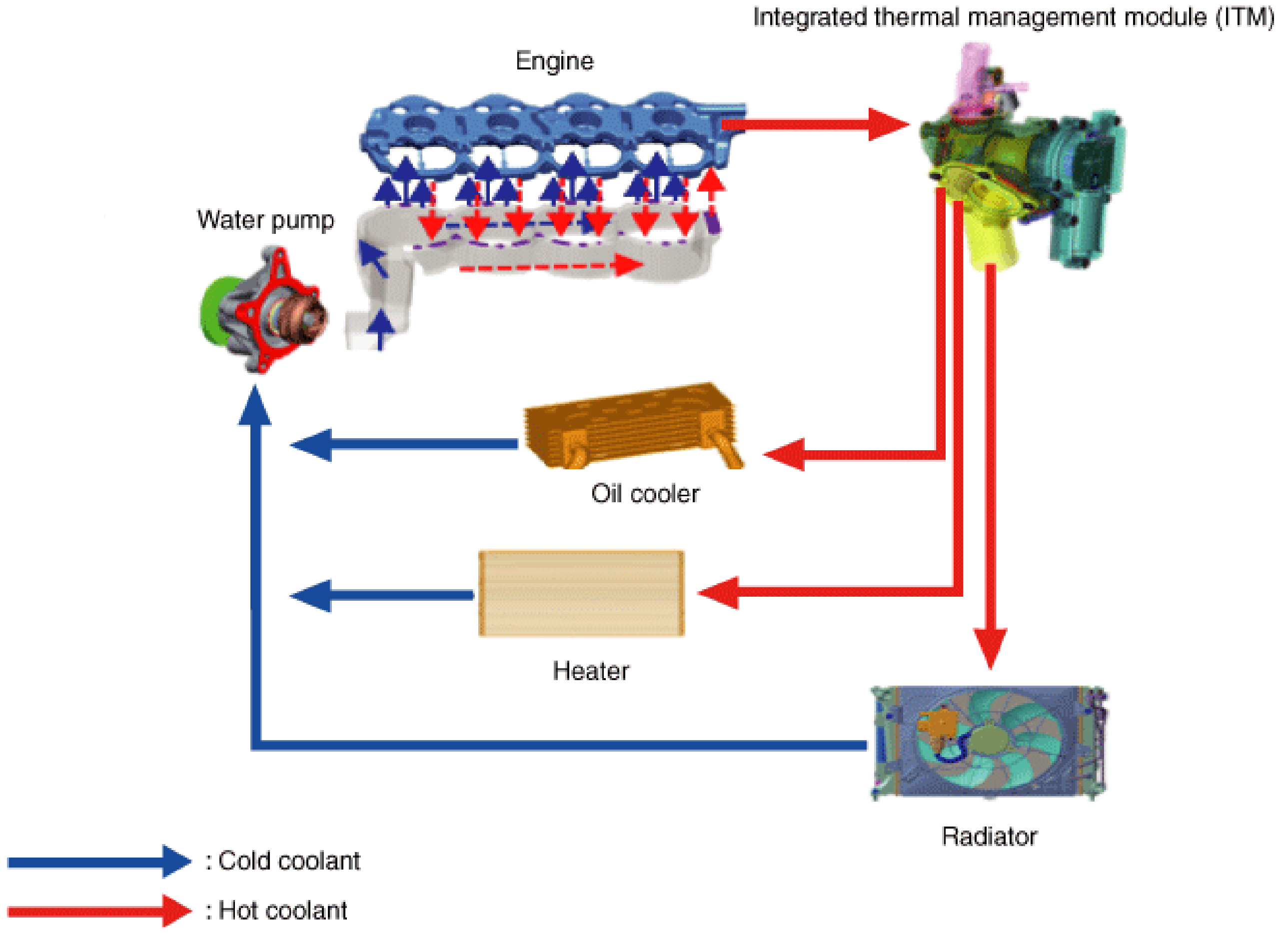
The valve uses seven different operating modes

– Mode 1: zero flow
– Mode 2: cabin heating
– Mode 3: cabin heating + by-pass
– Mode 4: cabin heating + engine oil heating
– Mode 5: engine oil heating
– Mode 6: engine oil cooling
– Mode 7: cooling after shutdown




X50A Engine Wiring Harness Junction Block
KR75 Engine Controls Ignition Relay
F88UA 15A
X3
H3
5294 VT/BU
J168
X966
3
IGN
A90 Logic
Q97B Engine Coolant Flow Control Valve - Block
Q97M Engine Coolant Flow Control Valve - Main
Q74 Engine Coolant Bypass Valve
Q97 Engine Coolant Flow Control Valve
M
2732 GN/BN
6550 BK
Data Communication Schematics
G176A
480 WH/RD
2973 GY
580 BK/GN
2971 GY/BU
2972 GY/BN
2978 BU/GY
2976 BU
2977 BU/BN
J180
Engine Controls Schematics
J169
X1
48
11
89
88
19
36
X2
84
83
5V2
K20 Engine Control Module

Hyundai: thermal management module



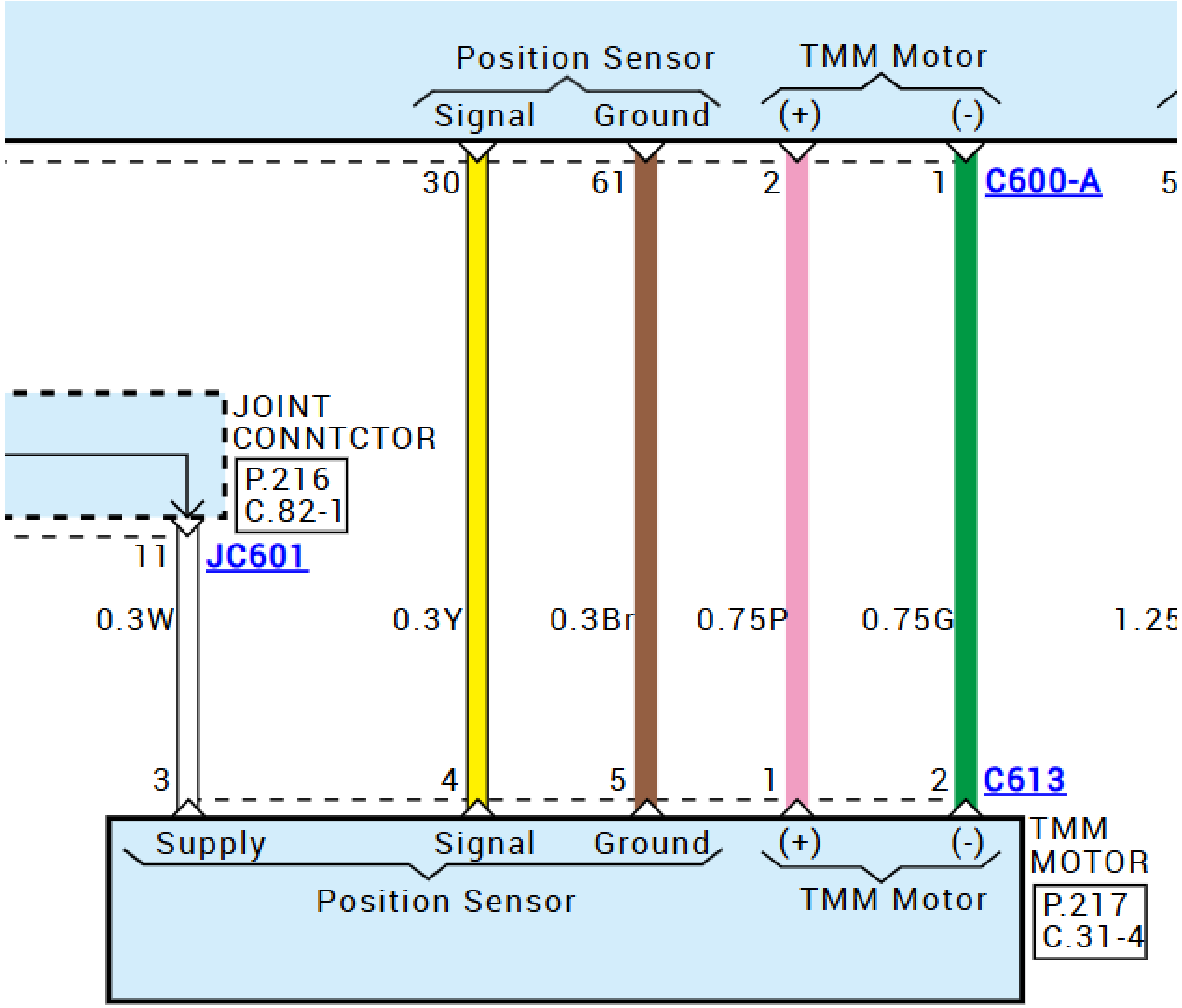
Hyundai: thermal management module

Supply: 5v

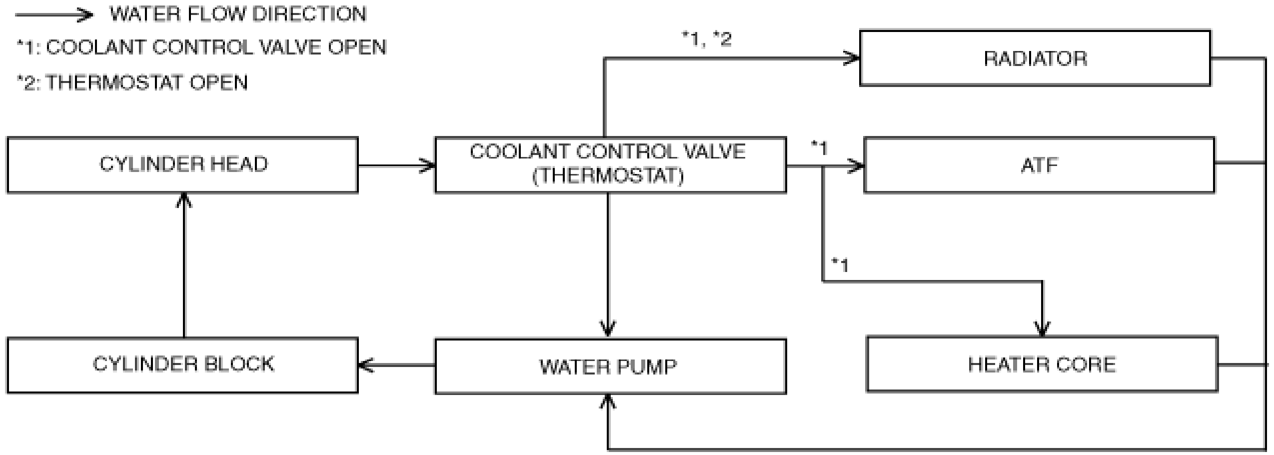
Signal position

Ground

Motor + - : 0-12V

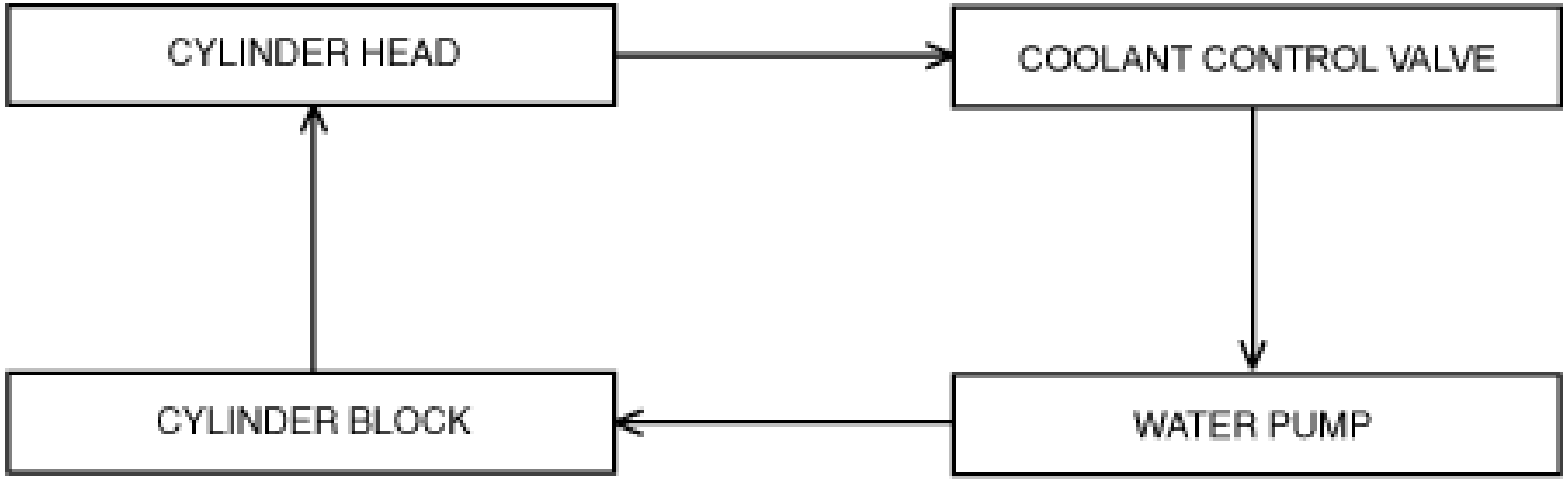


Mazda

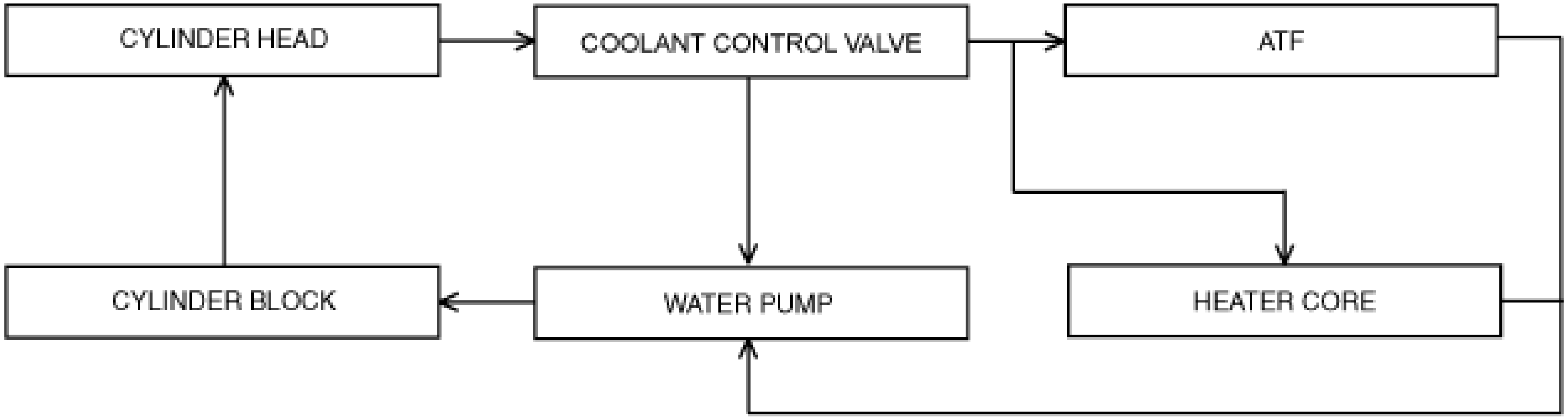

WATER FLOW DIRECTION
*1: COOLANT CONTROL VALVE OPEN
*2: THERMOSTAT OPEN
*1, *2
RADIATOR
CYLINDER HEAD
COOLANT CONTROL VALVE (THERMOSTAT)
*1
ATF
*1
CYLINDER BLOCK
WATER PUMP
HEATER CORE


FD

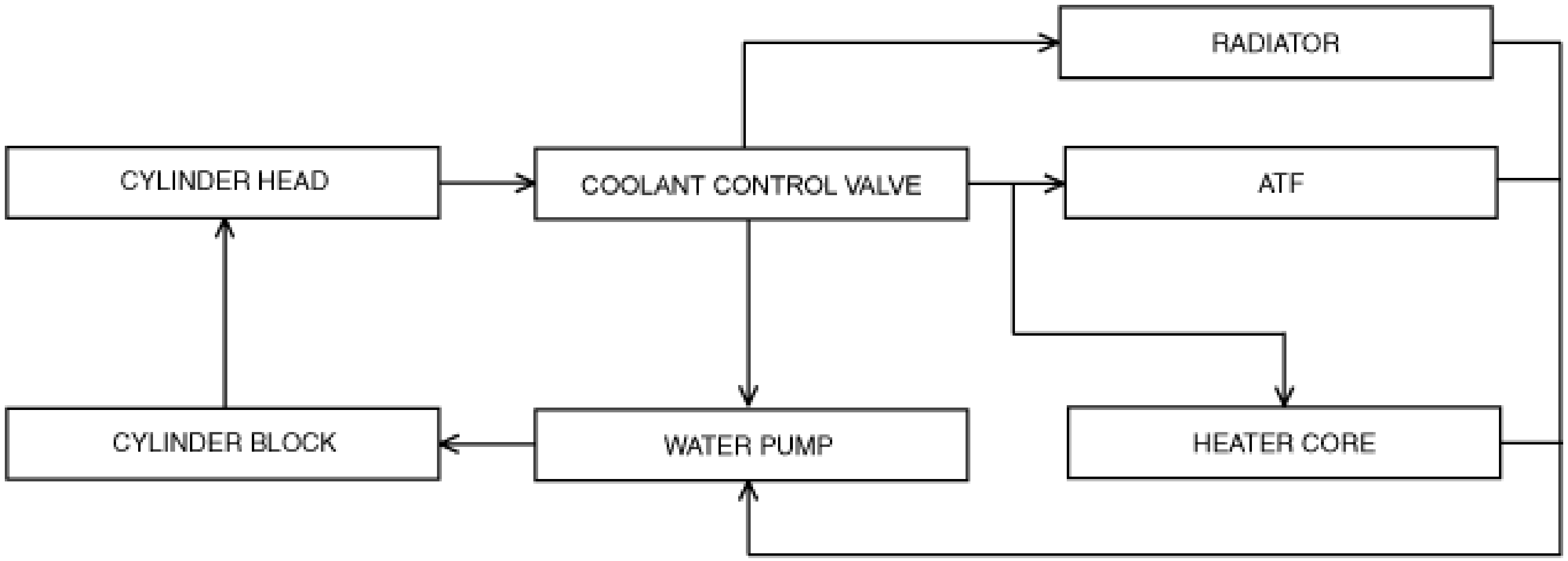
Flow constant

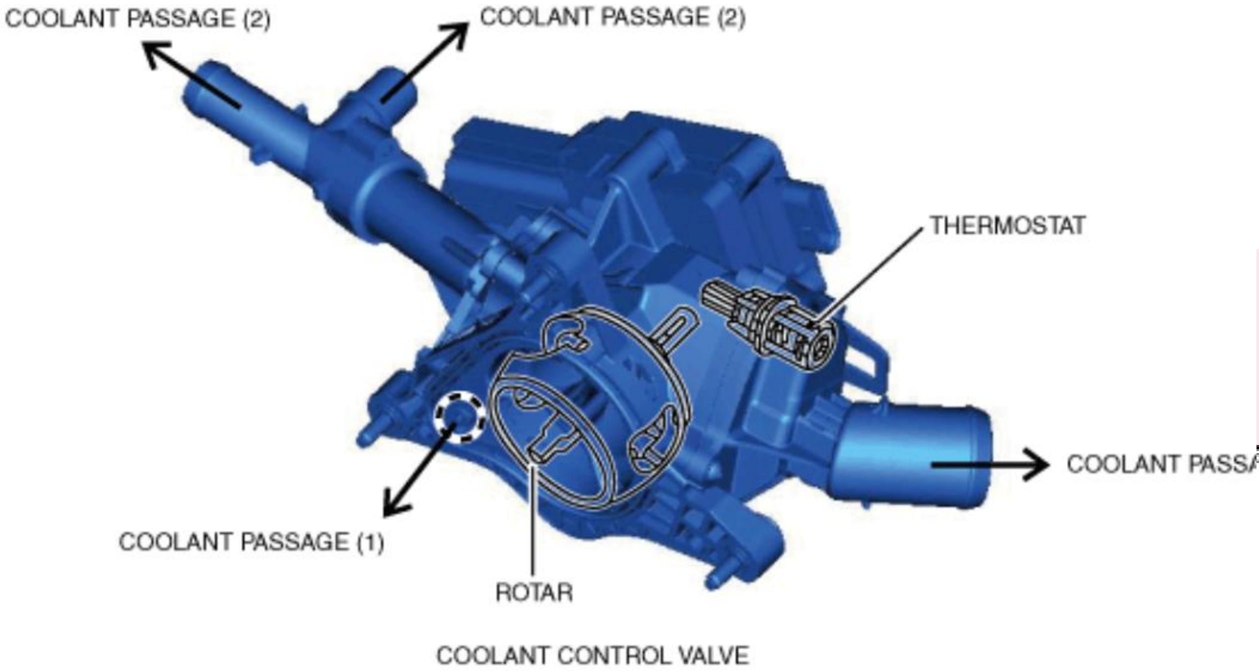


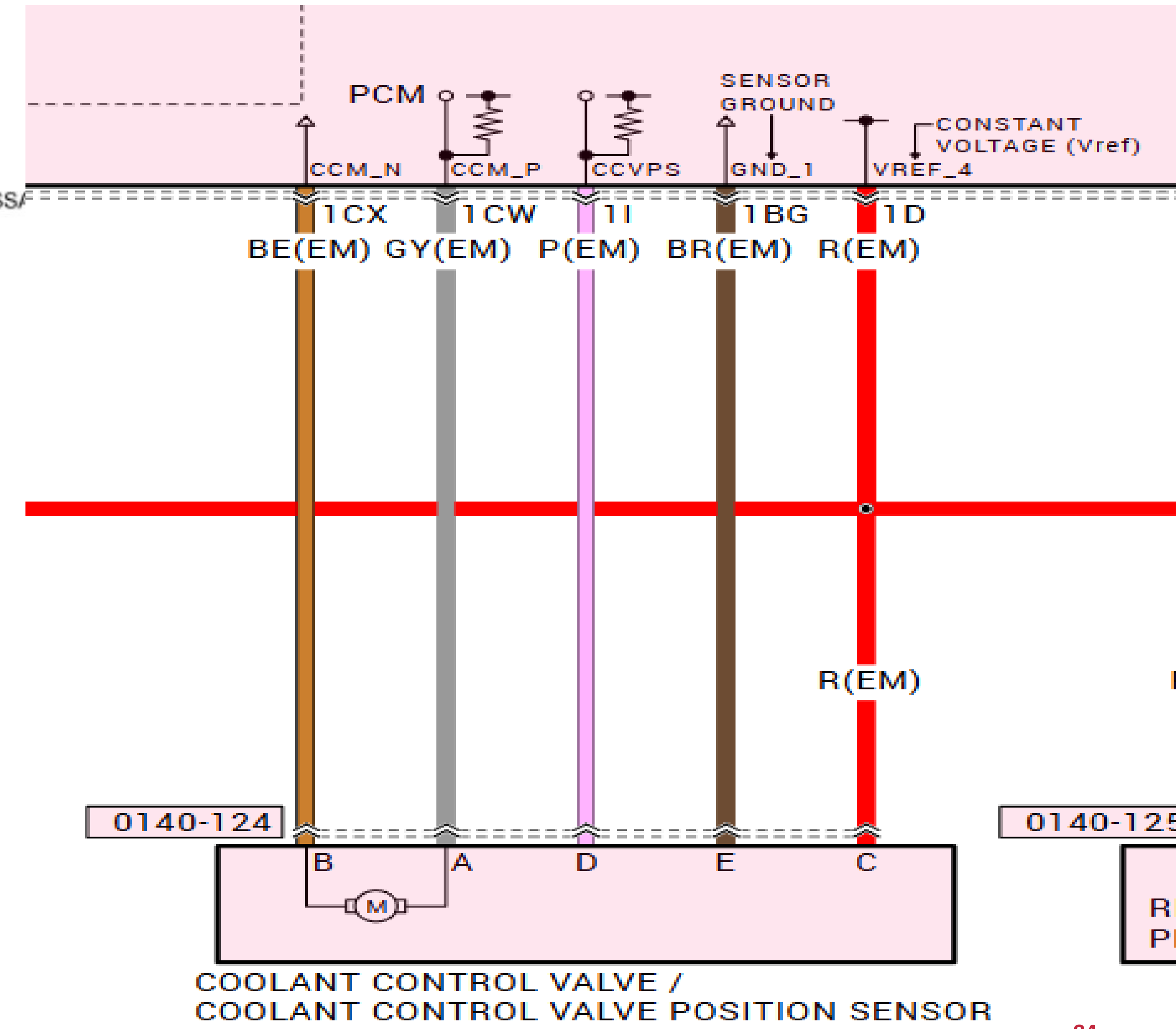
Condition before engine warm-up



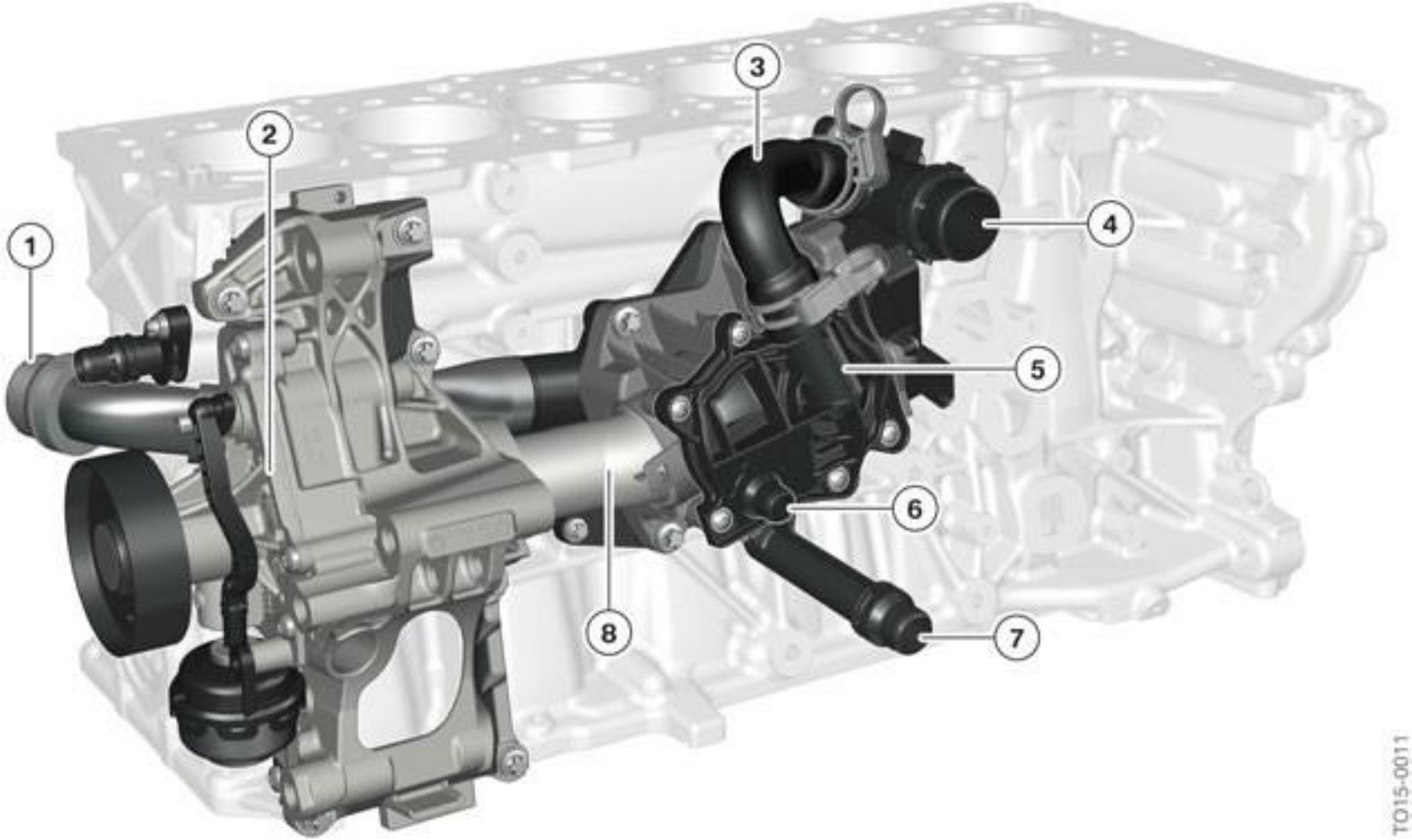
Engine at operating temperature



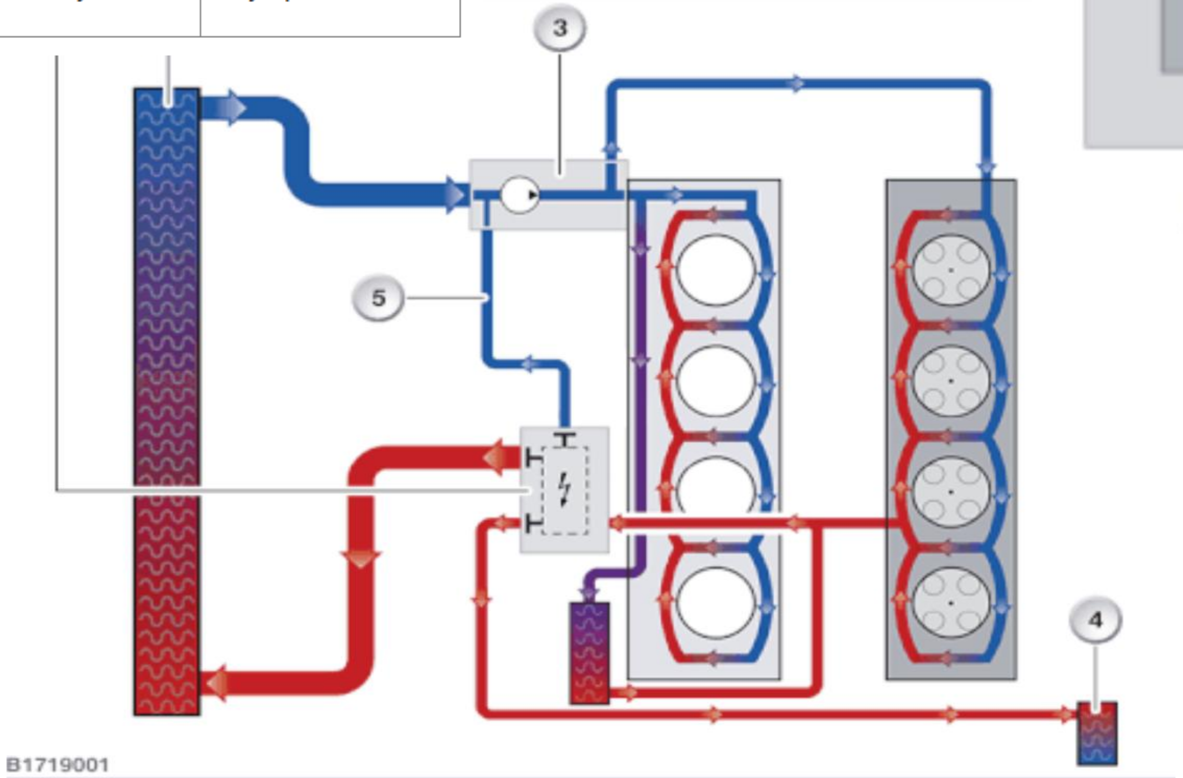

COOLANT PASSAGE (2)
COOLANT PASSAGE (2)
THERMOSTAT
COOLANT PASSAGE (1)
ROTAR
COOLANT CONTROL VALVE

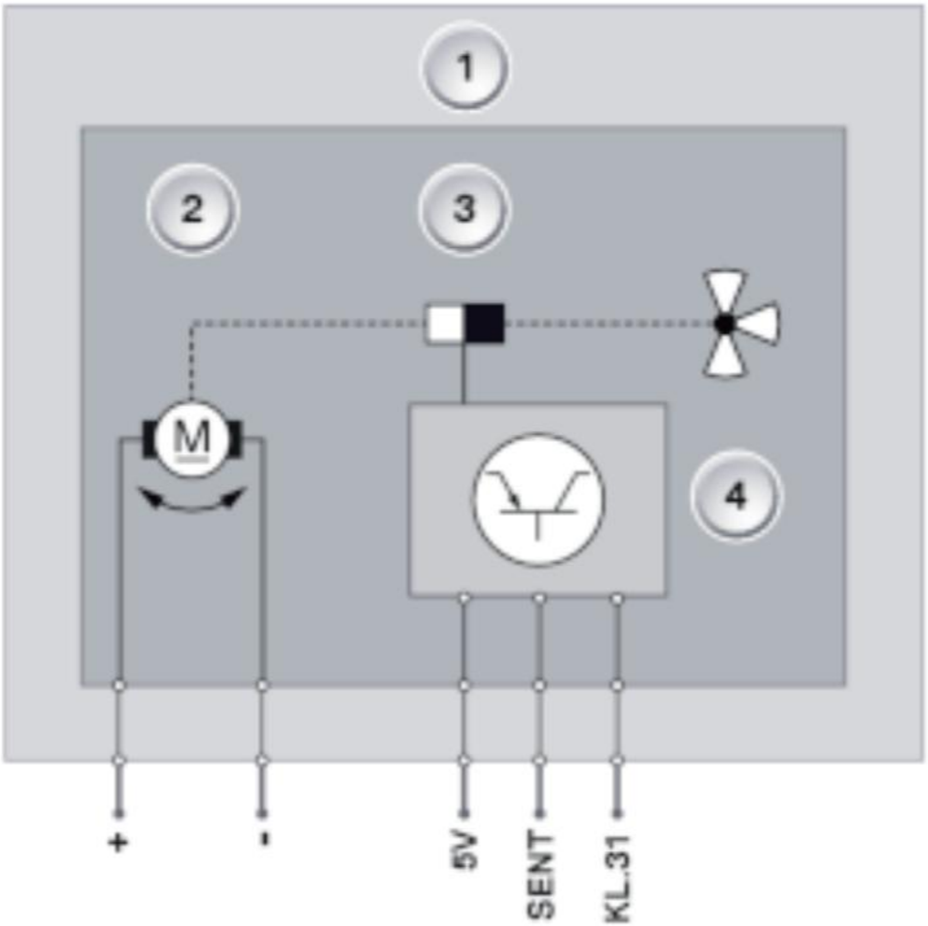

PCM
SENSOR GROUND
CONSTANT VOLTAGE (Vref)
CCM_N
CCM_P
CCVPS
GND_1
VREF_4
1CX
1CW
1I
1BG
1D
BE(EM)
GY(EM)
P(EM)
BR(EM)
R(EM)
R(EM)
0140-124
B
A
D
E
C
M
COOLANT CONTROL VALVE /
COOLANT CONTROL VALVE POSITION SENSOR

BMW heat management module



Activation	Valve position of bypass to the coolant pump	Valve position of heat exchanger for heating system	Valve position of coolant radiator
0 %	Fully closed	Fully closed	Fully closed
5%	Partially open	Fully closed	Fully closed
6 %	Partially open	Slightly open	Fully closed
10 %	Fully open	Fully open	Fully closed
11 %	Fully open	Fully open	Slightly open
90 %	Almost fully closed	Fully open	Fully open
100 %	Fully closed	Almost fully closed	Fully open





Coolant flush

Important!

The rules

– Never reuse old antifreeze
– Use a vac-n-fill
– Always follow the manufacturer's procedures

Coolant flush

The procedure

The steps

– Drain the system through the radiator drain if possible, or through a hose, the way the manufacturer recommends
– Refill the system with the correct coolant required by the manufacturer, according to the specification
– Bleed the whole system according to the manufacturer

KEY POINT — Most important: reprogram or calibrate the component you replaced.

7. Press the accelerator and brake pedal for two seconds at the same time, then release.

   Note:

   - There will NOT be a message on the Driver Information Center (DIC) to indicate that the ACSF is activated.
   - It may take up to 60 seconds for the ACSF procedure to become audible.

8. Start and run the engine at 1200 RPM for three minutes while adding coolant mixture as the level drops below the fill mark on the radiator surge tank until the level stabilizes.
9. Install the radiator surge tank cap.
10. Run the engine as follows while watching the coolant temperature gauge ensuring that the temperature is rising, but not overheating. If the radiator surge tank empties during the procedure below, stop the process and fill the radiator surge tank with the coolant mixture to the fill mark and start the procedure over.
    1. Run the engine speed at 3500 RPM for three minutes.
    2. Lower the engine speed to 1200 RPM and run for three minutes.
    3. Raise the engine speed to 3500 RPM and run for three minutes.
    4. Lower the engine speed to 1200 RPM and run for three minutes.

    Note: If you are unable to maintain the coolant level at the cold fill line of the surge tank and the surge tank empties while the process is running, turn the ignition off. The vehicle will only allow one ACSF process per key cycle.

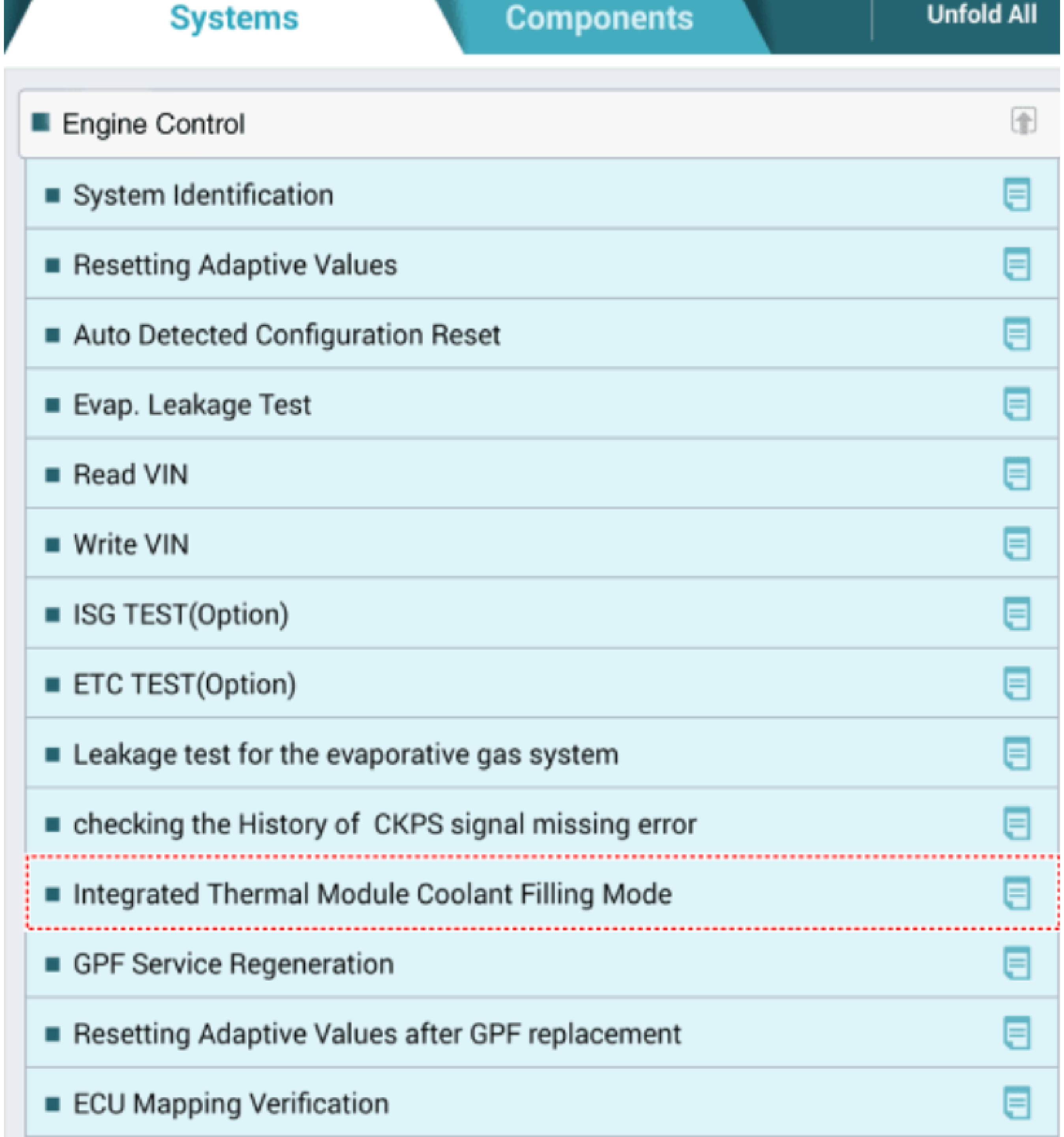
    1. Fill the radiator surge tank to the cold fill line and wait two minutes.
    2. Repeat steps 4-10 until the ACSF cycle completes with the coolant level maintained in the surge tank for the entire process.
11. Turn ignition off.
12. Allow the engine to cool
13. Adjust the coolant level in the surge tank to the cold fill line.
14. Install the radiator surge tank cap.
15. Start the engine and check for coolant leaks.

Vehicles on which the bleeding procedure is stored in the engine control module:

- If the vehicle has a parking/auxiliary heater, switch it on for about 30 seconds.
- Connect the Vehicle Diagnostic Tester .
- Start the program 01 Coolant Circuit Bleeding using the → Vehicle Diagnostic Tester and follow the instructions in the vehicle diagnostic tester display.

8. Start the engine and warm up the engine by slightly raising the engine speed above its normal idle speed until the cooling fan(s) comes on twice. Do not operate the air conditioning system.

9. Turn off the engine. Check the level in the tank, and add Honda Long Life Antifreeze/Coolant Type 2, if needed.


Systems
Components
Unfold All
Engine Control
System Identification
Resetting Adaptive Values
Auto Detected Configuration Reset
Evap. Leakage Test
Read VIN
Write VIN
ISG TEST(Option)
ETC TEST(Option)
Leakage test for the evaporative gas system
checking the History of CKPS signal missing error
Integrated Thermal Module Coolant Filling Mode
GPF Service Regeneration
Resetting Adaptive Values after GPF replacement
ECU Mapping Verification

Closing words!

Thank you all for taking part!

TRAINERS: Wilson Almeida

walmeida@vastauto.com